Pets4Life Project

I3 Deliverable

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Version 1.0

1. System Requirements

To ensure the system meets the goals outlined in the vision document, it must implement specific features that address both user and stakeholder needs. These are divided into functional and non-functional requirements. Functional requirements define what the system must do, such as enabling user registration or processing donations. Non-functional requirements describe how the system should perform, focusing on aspects like security, usability, and performance.

Each functional use case is also assigned a risk level—Low, Medium, or High—to indicate the potential impact of failure or misuse. High-risk use cases often involve sensitive data, transactions, or account management, while low-risk ones typically support general content or administrative tasks. Highlighting these risks helps guide development focus, security planning, and testing priorities.

1.1 Functional Requirements

Client and Pet Onboarding

- **ID: SR001 -** Enable new clients to register, creating profiles for themselves and their pets.
- **ID: SR002 -** Allow clients to upload veterinary records and other relevant pet information.
- **ID: SR003 -** Enable profile updates by clients, including contact information and pet details.
- **ID: SR004 -** Provide a system for password recovery and account security management.

Donation Management

- **ID: SR005 -** Process and record one-time and recurring donations.
- **ID: SR006 -** Automatically generate and send digital receipts for donations.
- **ID: SR007 -** Allow donors to select specific programs or funds they wish to support.
- **ID: SR008 -** Provide administrative tools for tracking and analyzing donation patterns.

Volunteer Management

- **ID: SR009 -** Enable volunteer registration with detailed profile creation.
- **ID: SR010 -** Calendar integration for scheduling volunteer activities.

- **ID: SR011** - Automated notifications for volunteer event reminders.
- **ID: SR012** - System for tracking volunteer hours and contributions.

Event Management

- **ID: SR013** - Allow staff to create and manage public events, including details such as date, time, location, and participant limits.
- **ID: SR014** - Online registration and payment for events.
- **ID: SR015** - Automated confirmation and reminder emails to event participants.
- **ID: SR016** - Features to promote events on social media and other platforms.

Pet Adoption Process

- **ID: SR017** - Display detailed profiles of sponsorable pets, including photos and health information.
- **ID: SR018** - Administrative interface for managing adoption applications and status updates.
- **ID: SR019** - Automated notifications to applicants about their application status.

Customer Support

- **ID: SR020** - Chatbot support for immediate assistance.
- **ID: SR021** - Ticketing system for handling and tracking customer inquiries.
- **ID: SR022** - FAQ section for common queries and concerns.
- **ID: SR023** - Feedback system for customers to rate services and provide suggestions.

Content Management

- **ID: SR024** - Tools for staff to post updates, blogs, and pet care information.
- **ID: SR025** - Photo and video gallery management for event and pet highlights.
- **ID: SR026** - Secure management of user-generated content, including reviews and comments.

- **ID: SR027** - Editorial tools for content revision and approval.

Administrative and Reporting Tools

- **ID: SR028** - Comprehensive dashboards for monitoring website traffic and user activity.
- **ID: SR029** - Detailed reporting tools for financial management, including donations and event revenues.

Use Case Scenarios

Based on the defined system requirements, we have identified 29 use cases that outline how various users will interact with the system to complete key tasks such as registering pets, managing donations, scheduling volunteers, and facilitating adoptions. These use cases help define the scope of the system and ensure that all functional needs are clearly mapped.

Each use case includes the use case name, primary actor, a brief description, and an assigned risk level. Following UML and Unified Process principles, a use case represents a goal that an actor wants to achieve through the system. As such, every use case is initiated by an actor and described from the actor's perspective.

The risk level for each use case was determined based on factors such as data sensitivity, potential for misuse, and system exposure. These classifications help prioritize development efforts and ensure appropriate safeguards are in place, as detailed in the Risk Assessment section.

1.2 **Use Case Overview**

Use Case 1: Client Registration

- **Primary Actor**: Pet Owner
- **Brief Description**: Allows new clients to create profiles for themselves and their pets.
- **Risk Level:** Low

Use Case 2: Upload Veterinary Records

- **Primary Actor**: Pet Owner
- **Brief Description**: Enables clients to upload veterinary records and other relevant pet information.
- **Risk Level:** High

Use Case 3: Profile Management

- **Primary Actor**: Pet Owner
- **Brief Description**: Allows clients to update their profiles, including contact information

and pet details.
- **Risk Level:** Low

Use Case 4: Password Recovery

- **Primary Actor**: Pet Owner
- **Brief Description**: Provides a system for password recovery and account security management.
- **Risk Level:** High

Use Case 5: Process Donations

- **Primary Actor**: Donor
- **Brief Description**: Allows donors to make one-time and recurring donations through the system.
- **Risk Level:** High

Use Case 6: Generate Donation Receipts

- **Primary Actor**: System
- **Brief Description**: Automatically generates and sends digital receipts for donations.
- **Risk Level:** Low

Use Case 7: Donation Allocation

- **Primary Actor**: Donor
- **Brief Description**: Enables donors to select specific programs or funds they wish to support.
- **Risk Level:** Low

Use Case 8: Donation Analytics

- **Primary Actor**: Fundraising Manager
- **Brief Description**: Provides administrative tools for tracking and analyzing donation patterns.
- **Risk Level:** Low

Volunteer Management

Use Case 9: Volunteer Registration

- **Primary Actor**: Volunteer
- **Brief Description**: Enables volunteers to register with detailed profile creation.

- **Risk Level:** Low

Use Case 10: Volunteer Scheduling

- **Primary Actor**: Volunteer Coordinator
- **Brief Description**: Integrates a calendar system for scheduling volunteer activities.
- **Risk Level:** Low

Use Case 11: Volunteer Notifications

- **Primary Actor**: System
- **Brief Description**: Sends automated notifications for volunteer event reminders.
- **Risk Level:** Low

Use Case 12: Volunteer Tracking

- **Primary Actor**: Volunteer Coordinator
- **Brief Description**: Tracks volunteer hours and contributions for record-keeping.
- **Risk Level:** Low

Event Management

Use Case 13: Event Creation

- **Primary Actor**: Staff Member
- **Brief Description**: Allows staff to create and manage public events with relevant details.
- **Risk Level:** Low

Use Case 14: Event Registration & Payment

- **Primary Actor**: Pet Owner / Donor
- **Brief Description**: Enables users to register and make payments for events online.
- **Risk Level:** High

Use Case 15: Event Notifications

- **Primary Actor**: System
- **Brief Description**: Sends automated confirmation and reminder emails to event participants.
- **Risk Level:** Low

Use Case 16: Event Promotion

- **Primary Actor**: Marketing Staff
- **Brief Description**: Provides tools to promote events via social media and other platforms.
- **Risk Level:** Low

Pet Adoption Process

Use Case 17: Display Pet Profiles

- **Primary Actor**: Public User
- **Brief Description**: Displays detailed profiles of available pets for adoption.
- **Risk Level:** Low

Use Case 18: Adoption Management

- **Primary Actor**: Shelter Staff
- **Brief Description**: Enables staff to manage adoption applications and update status.
- **Risk Level:** High

Use Case 19: Adoption Notifications

- **Primary Actor**: System
- **Brief Description**: Sends automated notifications to applicants about their application status.
- **Risk Level:** Low

Customer Support

Use Case 20: Chatbot Assistance

- **Primary Actor**: Pet Owner
- **Brief Description**: Provides immediate customer support via chatbot.
- **Risk Level:** Low

Use Case 21: Ticketing System

- **Primary Actor**: Customer Support Staff
- **Brief Description**: Implements a ticketing system to track customer inquiries.
- **Risk Level: High**

Use Case 22: FAQ Section

- **Primary Actor**: Public User

- **Brief Description**: Provides a knowledge base for common queries and concerns.
- **Risk Level:** Low

Use Case 23: Customer Feedback

- **Primary Actor**: Pet Owner
- **Brief Description**: Allows customers to submit service feedback and ratings.
- **Risk Level:** Low

Content Management

Use Case 24: Publish Content

- **Primary Actor**: Staff Member
- **Brief Description**: Allows staff to post updates, blogs, and pet care information.
- **Risk Level:** Low

Use Case 25: Media Management

- **Primary Actor**: Staff Member
- **Brief Description**: Manages photo and video galleries for events and pet highlights.
- **Risk Level:** Low

Use Case 26: User-Generated Content

- **Primary Actor**: System Administrator
- **Brief Description**: Manages user-generated content, including reviews and comments.
- **Risk Level:** High

Use Case 27: Editorial Control

- **Primary Actor**: Content Manager
- **Brief Description**: Provides editorial tools for content revision and approval.
- **Risk Level:** Low

Administrative and Reporting Tools

Use Case 28: Website Analytics Dashboard

- **Primary Actor**: Administrator
- **Brief Description**: Monitors website traffic and user activity.
- **Risk Level:** Low

Use Case 29: Financial Reports

- **Primary Actor**: Administrator
- **Brief Description**: Generates reports for financial management, including donations and event revenues.
- **Risk Level:** High

Associated System Requirements and Traceability Matrix

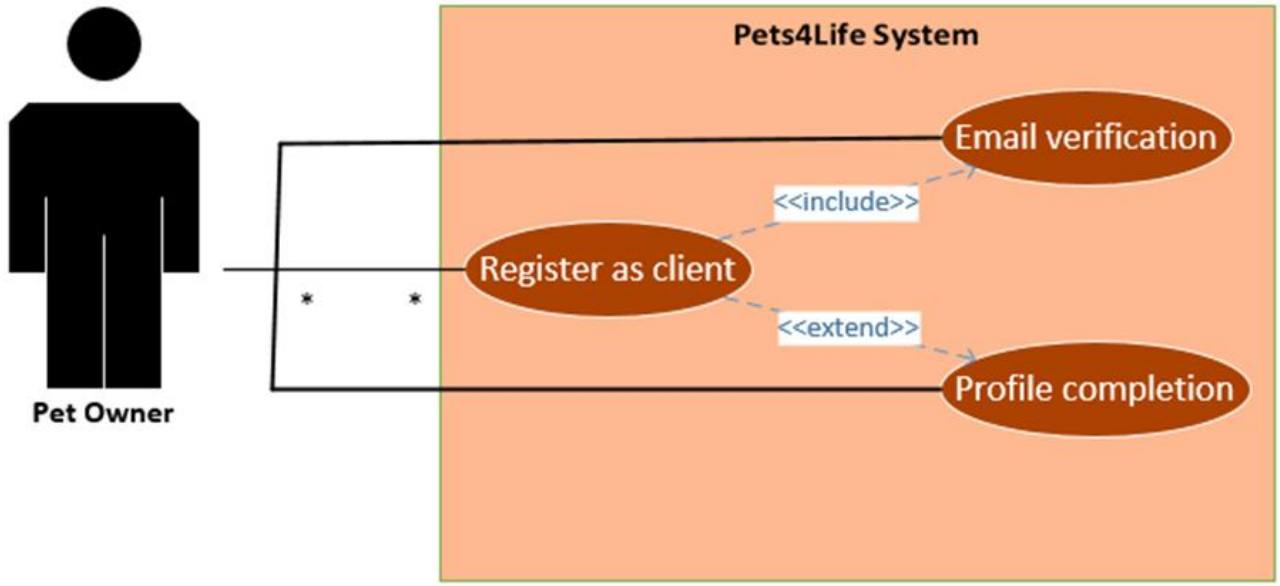
System Requirements:

- **SR001: The system will enable new clients to register, creating profiles for themselves and their pets**
- **SR002: The system will allow clients to upload veterinary records and other relevant pet information**
- **SR003: The system will enable profile updates by clients, including contact information and pet details**
- **SR004: The system will provide a system for password recovery and account security management**
- **SR005: The system will process and record one-time and recurring donations**
- **SR006: The system wil automatically generate and send digital receipts for donations**

Pets4Life	Use Case 1: Client Registration	Use Case 2: Upload Veterinary Records	Use Case 3: Profile Management	Use Case 4: Password Recovery	Use Case 5: Process Donations	Use Case 6: Generate Donation Receipts
D: SR001 - Enable new clients to register, creating profiles for themselves and their pets	☑					
D: SR002 - Allow clients to upload veterinary records and other relevant pet information		☑				
D: SR003 - Enable profile updates by clients, including contact information and pet details			☑			
D: SR004 - Provide a system for password recovery and account security management.				☑		
D: SR005 - Process and record one-time and recurring donations					☑	
D: SR006 - Automatically generate and send digital receipts for donations						☑

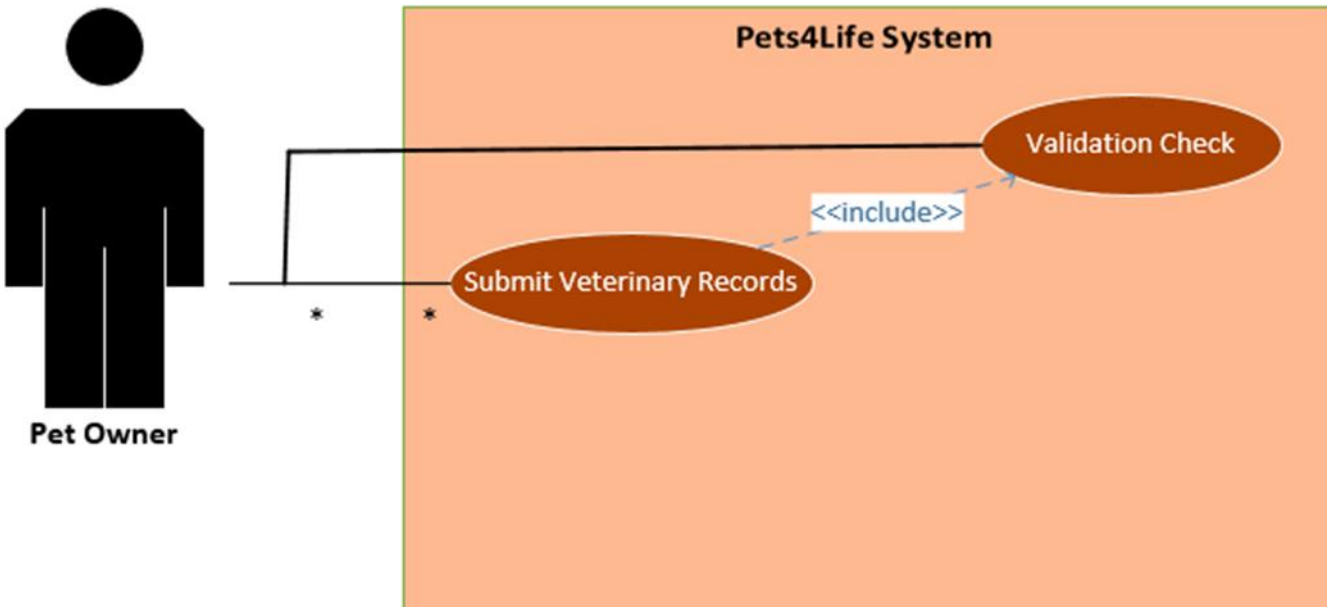
Use Case Diagrams 1-6

Use Case #1 Client Registration



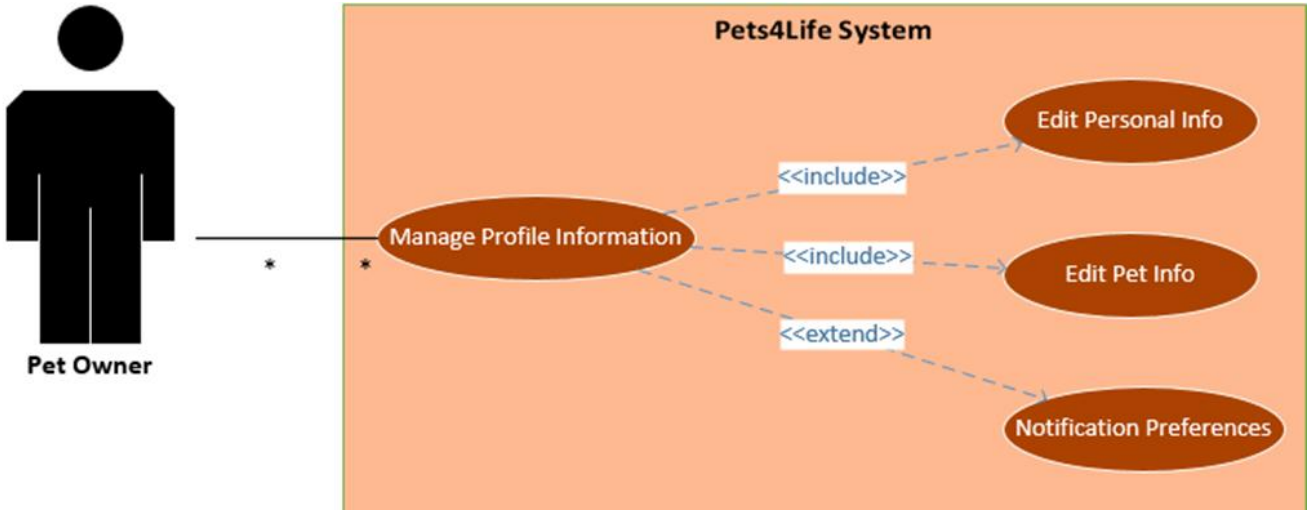
Use Case Diagram Narrative:
The Client Registration use case represents the process of a new pet owner signing up on the Pets4Life system. The Pet Owner initiates the registration, and the system requires Email Verification to confirm their identity. Once verified, the registration process extends to Profile Completion, where the user can provide additional personal details and pet information. The use case ensures that every user has a validated and complete profile before accessing other system features.

Use Case #2 Upload Veterinary Records



Use Case Diagram Narrative:
The Upload Veterinary Records case outlines how a Pet Owner submits important veterinary information for their pet. Upon uploading the records, the system performs a Validation Check to ensure that the files meet format and data accuracy requirements. This ensures that all submitted medical records are properly stored and accessible for future reference, improving pet data management.

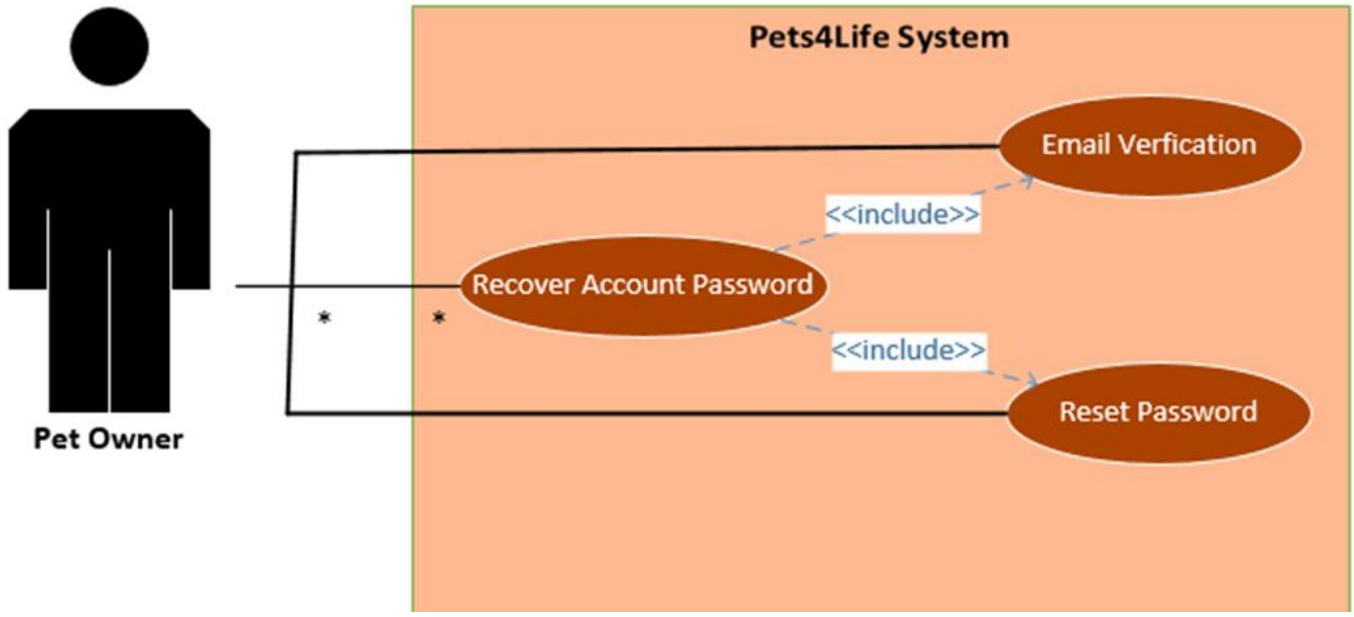
Use Case #3 Profile Management



Use Case Diagram Narrative:

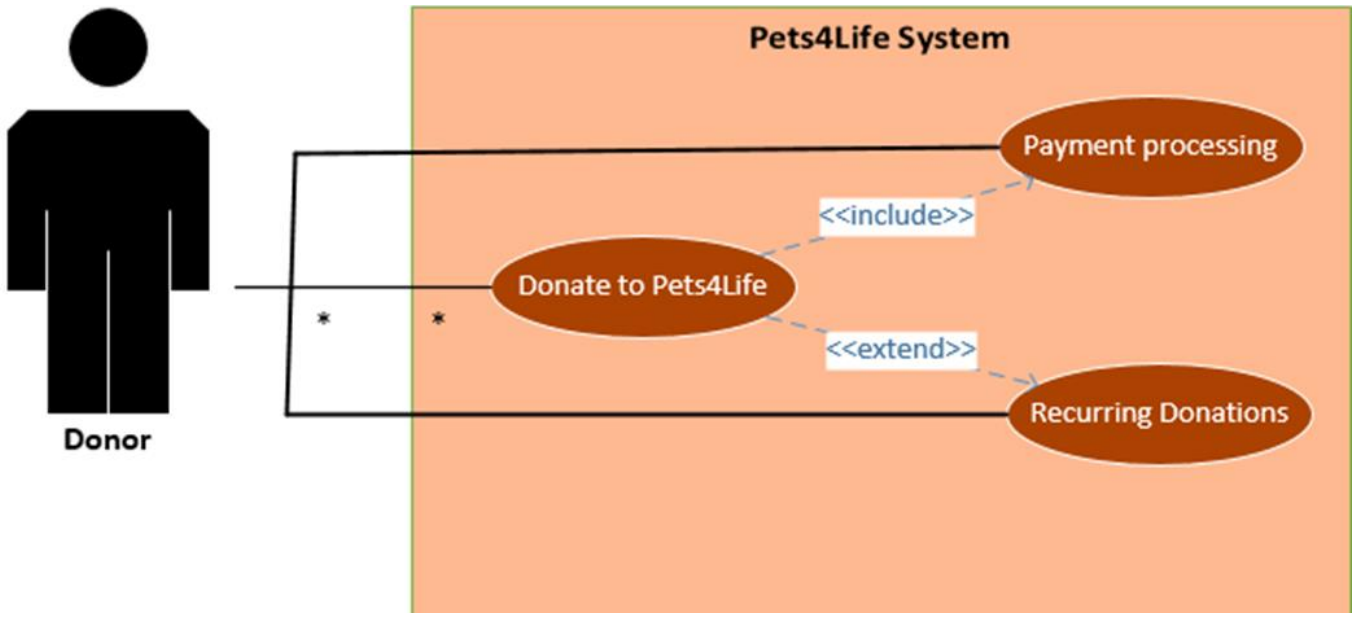
The Profile Management use case allows the Pet Owner to update and maintain their account information. The system includes options to Edit Personal Info and Edit Pet Info, ensuring that both user and pet details remain up to date. Additionally, the process extends to Notification Preferences, allowing users to customize the types of alerts and messages they receive from the system. This use case enhances user experience by providing flexible profile control.

Use Case #4 Password Recovery



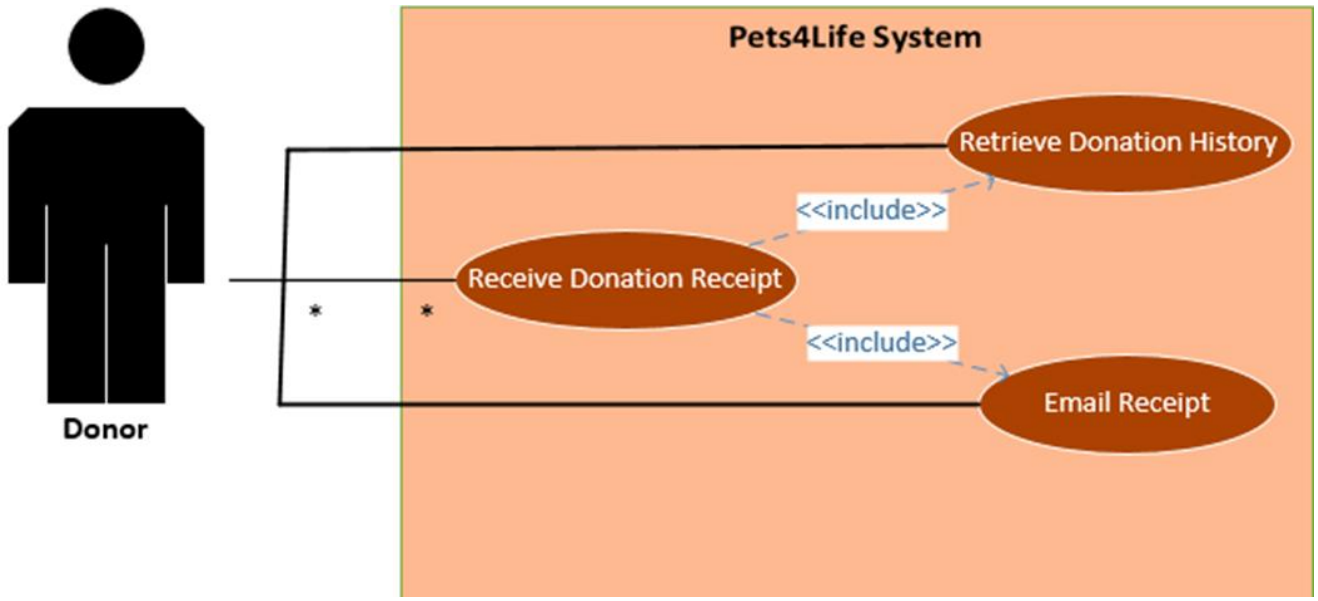
Use Case Diagram Narrative:
The Password Recovery use case provides a way for a Pet Owner to regain access to their account if they forget their login credentials. The recovery process begins with Email Verification, ensuring that the user is authorized to reset their password. Once verified, the system includes a Reset Password step, where the user can create a new secure password. This process strengthens account security and prevents unauthorized access.

Use Case #5 Process Donations



Use Case Diagram Narrative:
The Process Donations use case outlines how a Donor contributes to Pets4Life. The donation process includes Payment Processing to handle transactions securely. Additionally, the process extends to Recurring Donations, allowing donors to set up automatic contributions if they choose. This ensures that donations are managed efficiently, with options for both one-time and recurring support.

Use Case #6 Generate Donation Receipts



Use Case Diagram Narrative:
The Generate Donation Receipts use case describes how a Donor receives confirmation of their contribution. The system includes Retrieve Donation History, allowing donors to access past transactions, and Email Receipt, which sends an official confirmation to their inbox. This use case ensures that donors have proper documentation of their contributions and improves transparency within the donation system.

Use Cases 7-13

System requirements:

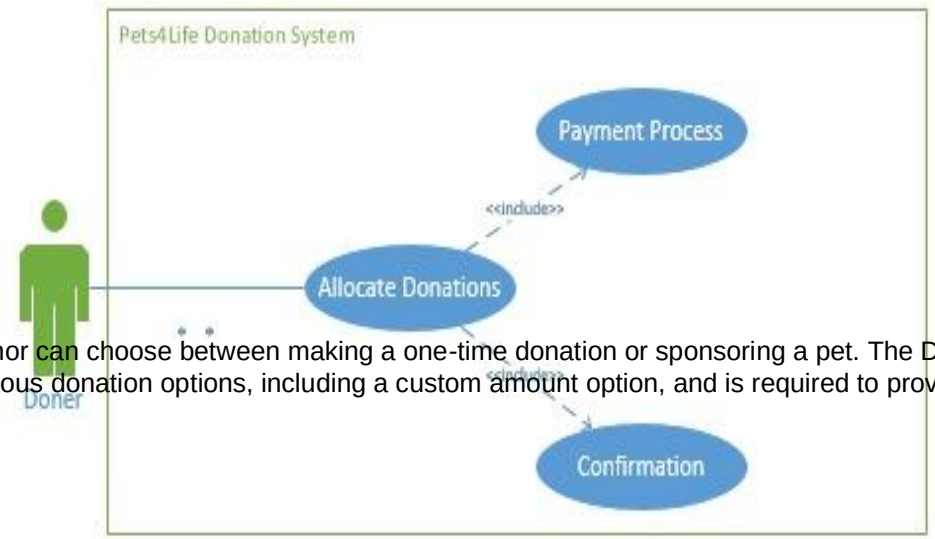
ID: SR007 – The system will allow donors to select specific programs or funds they wish to support.

ID: SR008 – The system will provide administrative tools for tracking and analyzing donation patterns

ID: SR009 – The system will enable volunteer registration with detailed profile creation.
ID: SR010 – The system will calendar integration for scheduling volunteer activities.
ID: SR011 – The system will have automated notifications for volunteer event reminders.
ID: SR012 – The system will track volunteer hours and contributions.

1	Pets4Life	Use Case 7: Donation Allocation	Use Case 8: Donation Analytics	Use Case 9: Volunteer Registration	Use Case 10: Volunteer Scheduling	Use Case 11: Volunteer Notifications	Use Case 12: Volunteer Tracking
2	ID: SR007 - Allow donors to select specific programs or funds they wish to support	☑					
3	ID: SR008 - Provide administrative tools for tracking and analyzing donation patterns		☑				
4	ID: SR009 - Enable volunteer registration with detailed profile creation			☑			
5	ID: SR010 - Calendar integration for scheduling volunteer activities				☑		
6	ID: SR011 - Automated notifications for volunteer event reminders					☑	
7	ID: SR012 - System for tracking volunteer hours and contributions						☑

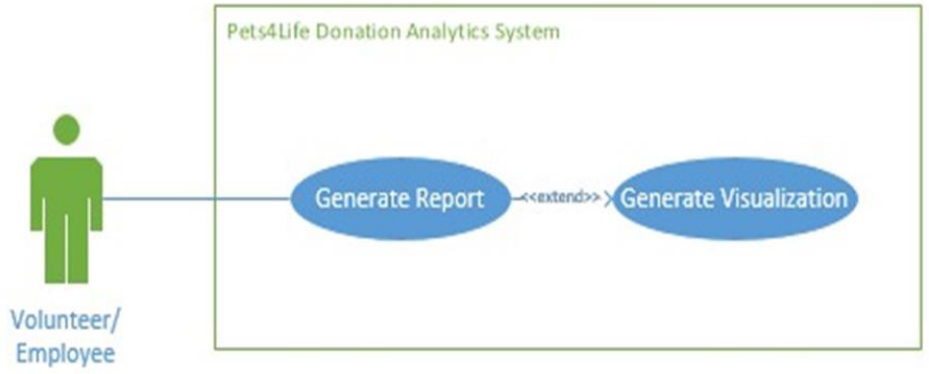
Use Case #7



The Donor can choose between making a one-time donation or sponsoring a pet. The Donor is presented with various donation options, including a custom amount option, and is required to provide personal

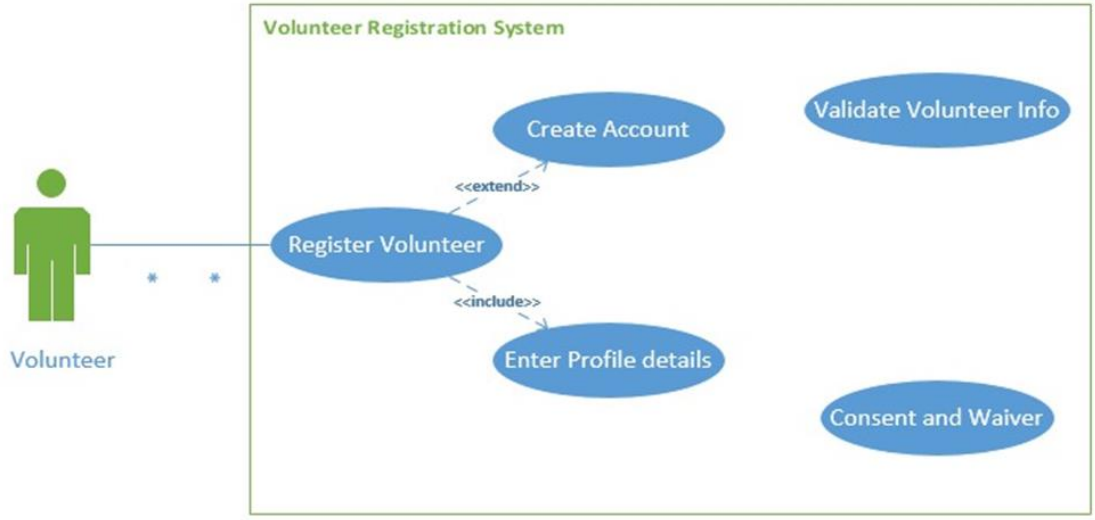
information (name, email, address) to proceed with the donation. Once the Donor submits the information, the system processes the donation and records the details for future reference. If the Donor chooses to sponsor a pet, the pet's sponsorship status is updated accordingly. The **Donor** is the parent actor, with the ability to either make a one-time donation or sponsor a pet, inheriting all actions such as selecting the donation type, entering personal details, and submitting the donation.

Use Case #8



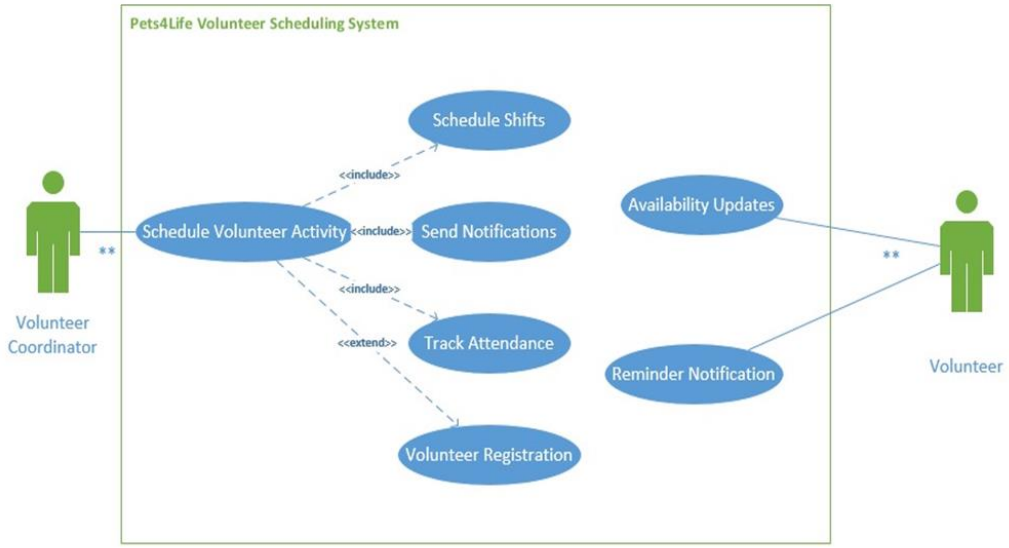
The Fundraising Manager can access analytical tools that allow them to track and analyze donation patterns. They can filter donations based on time period, donor type, or donation amount. The system provides visual reports, highlighting donation trends and comparisons, allowing the Fundraising Manager to make data-driven decisions for future campaigns. The **Fundraising Manager** is the parent actor and can access various analysis tools and reports, inheriting actions like filtering donations and viewing the reports for further decision-making.

Use Case #9



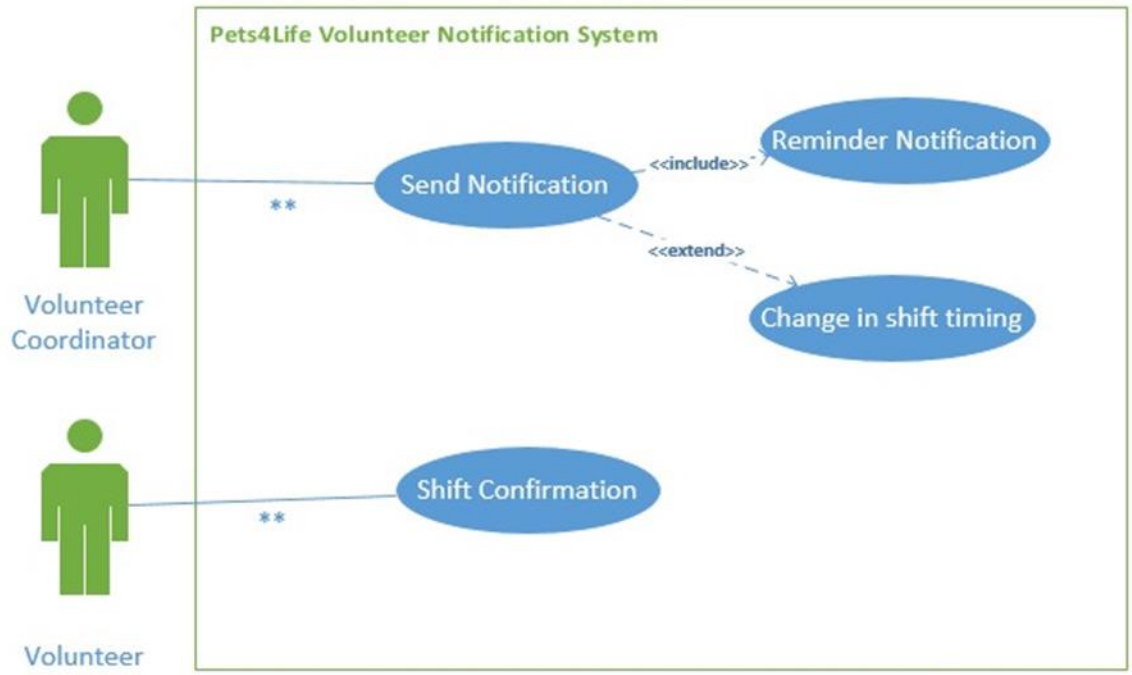
The Volunteer can register by creating a detailed profile within the system. This process requires the Volunteer to enter personal information, including their name, contact details, availability, and volunteer interests. Once the Volunteer completes their registration, their profile is stored in the system for event scheduling and communication purposes. The **Volunteer** is the parent actor and is responsible for registering, managing their profile, and receiving future event invitations, inheriting all actions associated with profile creation and event registration.

Use Case #10



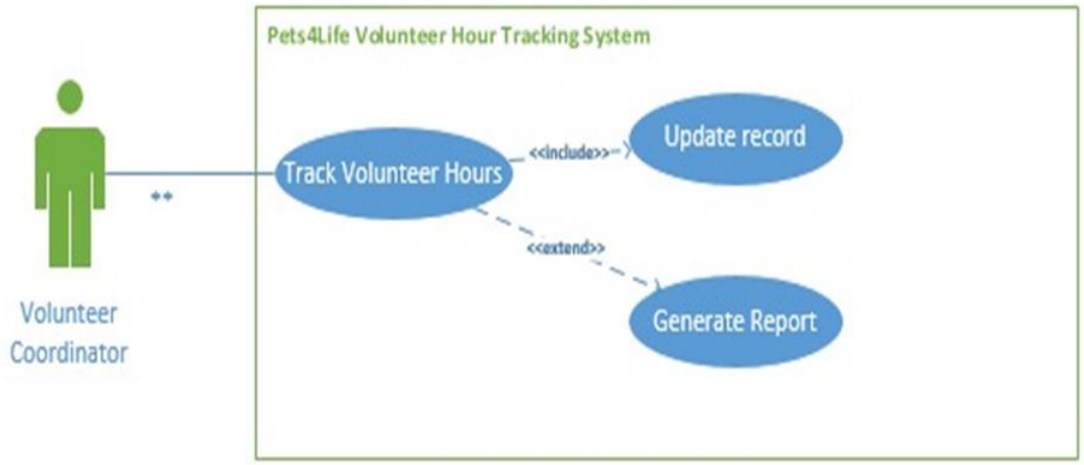
The Volunteer can view a list of available events and select dates they wish to attend, confirming their attendance within the system. The system then manages event scheduling, ensuring volunteers are assigned to appropriate shifts based on their availability. Additionally, Volunteers will receive notifications about upcoming events they have registered for. The **Volunteer** is the parent actor and can view and select event dates. The **System** manages event assignments and notifies the Volunteers, with actions related to event scheduling inherited by the **Volunteer**.

Use Case #11



The System is responsible for sending automated notifications to Volunteers about event reminders, shifts, and other important updates. Volunteers will be notified via email or SMS about upcoming volunteer opportunities, any changes to their scheduled events, or cancellations. The **System** is the primary actor, sending notifications to the **Volunteer**, and the Volunteer inherits actions like receiving and acknowledging notifications for upcoming events or updates.

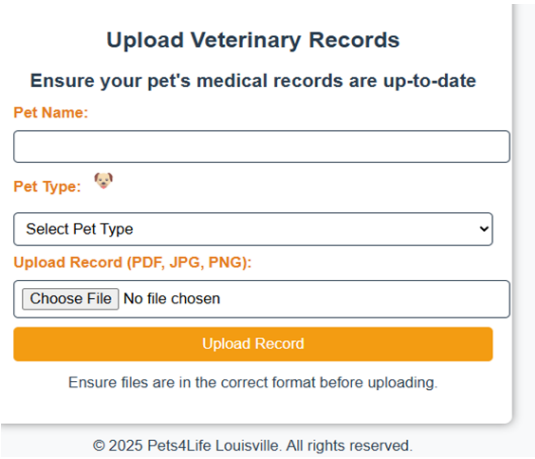
Use Case #12

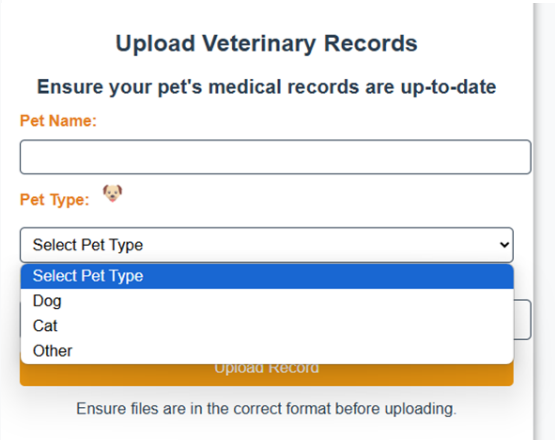


The Volunteer Coordinator is responsible for tracking volunteer hours and contributions across various events. The Coordinator logs the hours worked by each Volunteer and generates reports based on this data. The system stores these hours for record-keeping purposes, and the Volunteer Coordinator can access historical data to analyze volunteer activity and engagement. The **Volunteer Coordinator** is the parent actor, responsible for tracking and recording volunteer hours, while the **Volunteer** inherits actions related to logging and reviewing the hours worked, ensuring accurate and up-to-date record-keeping.

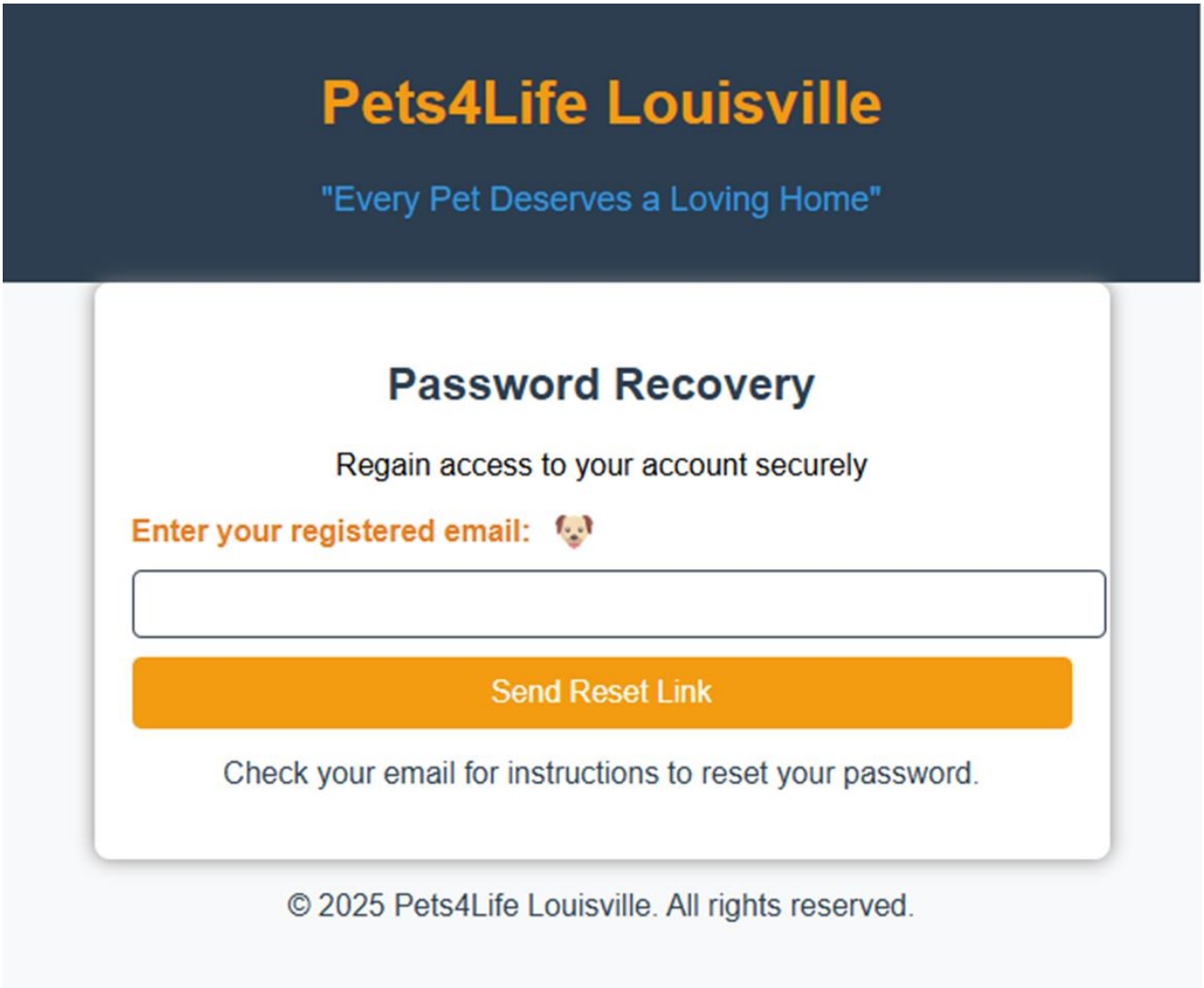
1.6 High Risk Prototypes:

Use Case 2: Upload Veterinary Records

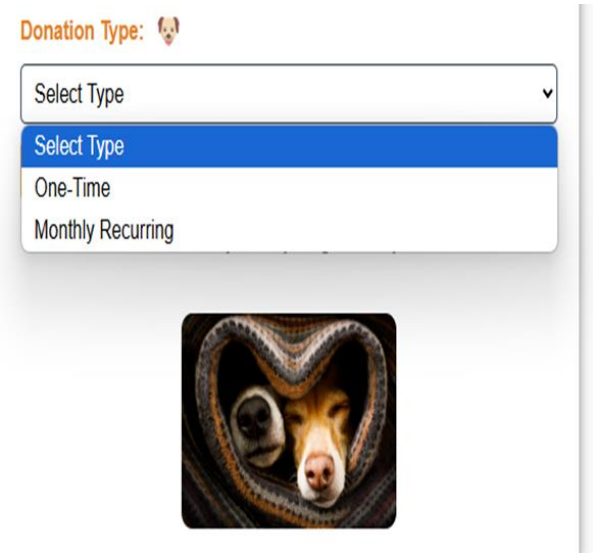


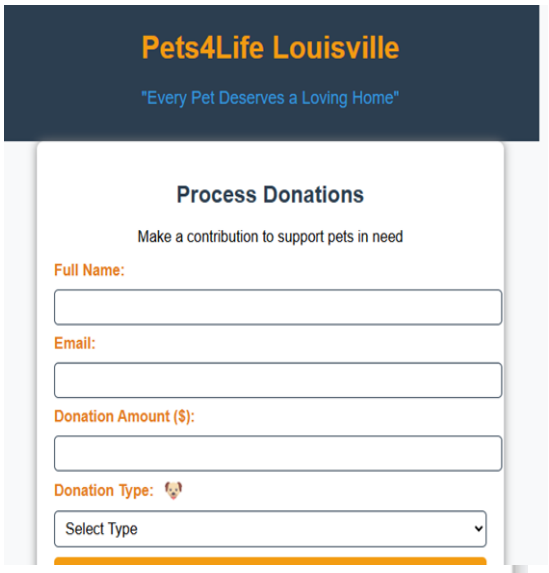


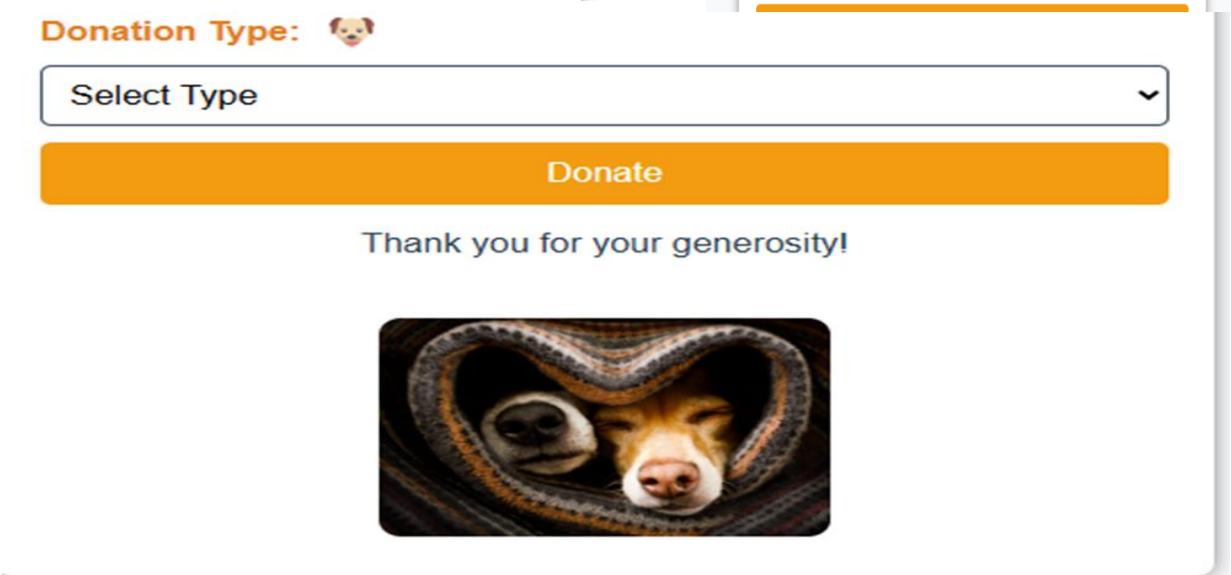
Use Case 4: Password Recovery



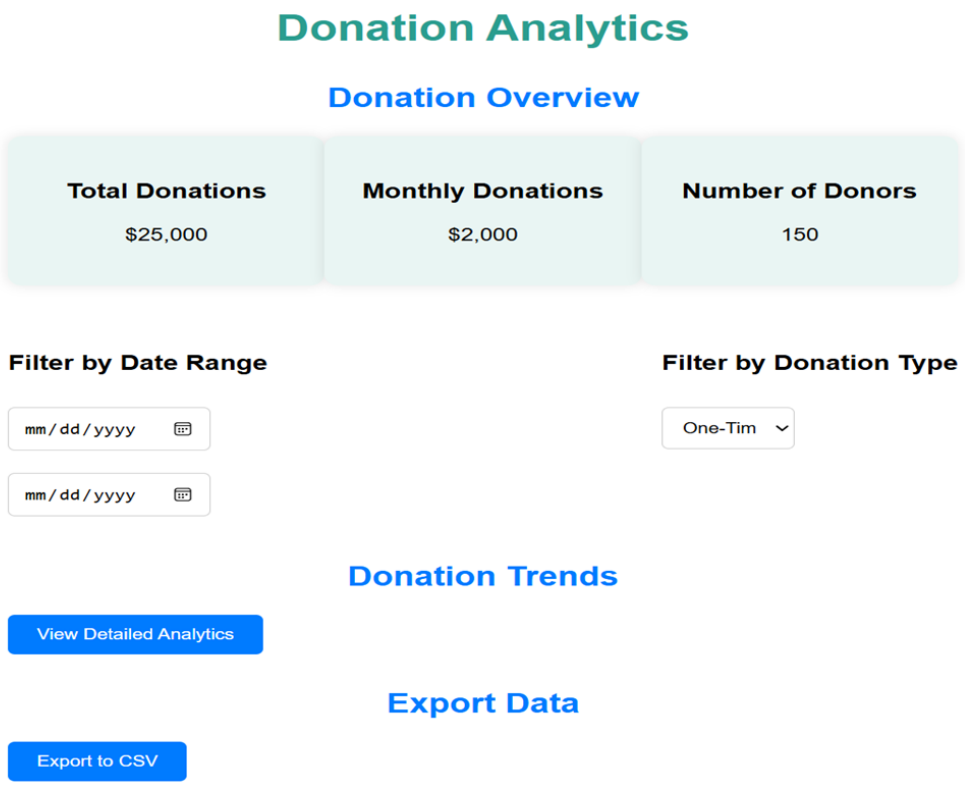
Use Case 5: Process Donations



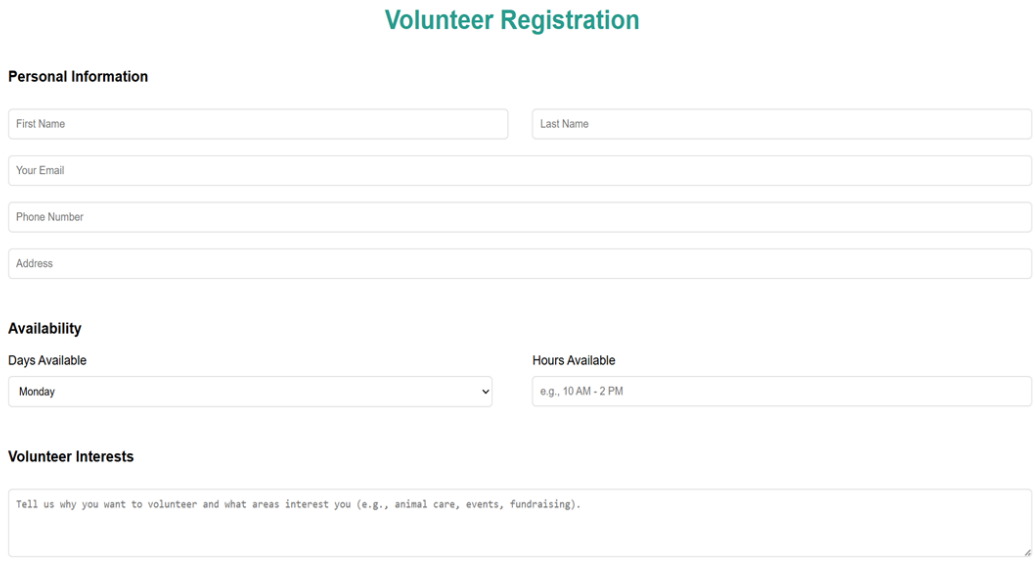




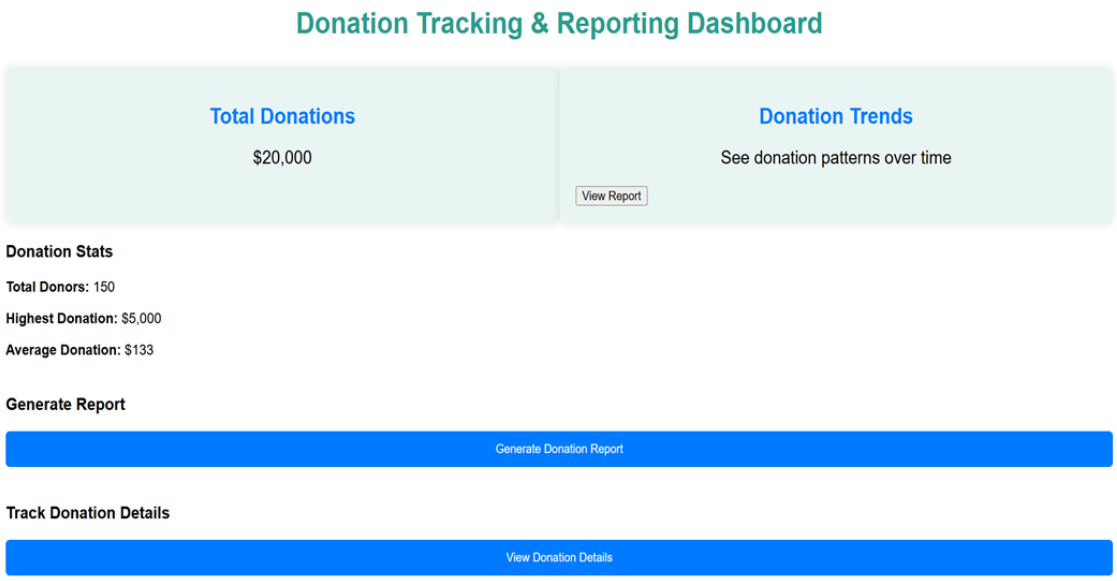
Use Case 8: Donation Analytics



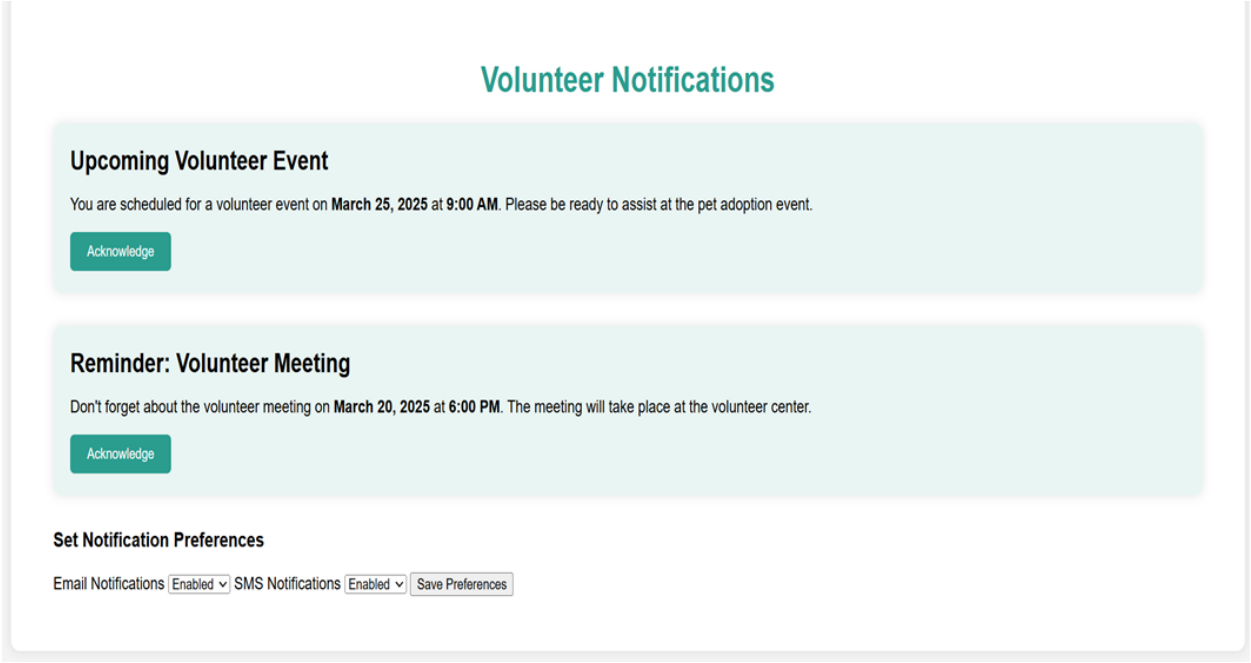
Use Case 9: Register Volunteer



Use Case 10: Schedule Volunteer Activities



Use Case 11: Send Volunteer Notifications



Use Case 12: Track Volunteer Hours and Contributions

Volunteer Tracking

Volunteer: John Doe

Email: john.doe@email.com

Phone: (555) 123-4567

Volunteer Role: Event Coordinator

Volunteer Hours Logged

Date	Event	Hours Worked	Notes
March 5, 2025	Pet Adoption Event	5	Assisted with event setup
March 10, 2025	Volunteer Orientation	2	Helped with training new volunteers
March 12, 2025	Fundraising Event	6	Coordinated donation collection

Log New Volunteer Hours

Event Name

mm/dd/yyyy

Volunteer Hours Logged

Date	Event	Hours Worked	Notes
March 5, 2025	Pet Adoption Event	5	Assisted with event setup
March 10, 2025	Volunteer Orientation	2	Helped with training new volunteers
March 12, 2025	Fundraising Event	6	Coordinated donation collection

Log New Volunteer Hours

Event Name

mm/dd/yyyy

Hours Worked

Notes (optional)

Log Hours

Use case 13-19

1.2.A System Requirements

- SR013: system will allow staff to create event listings.
- SR014: system will allow public registration for events.
- SR015: system will allow users to make payment and donation
- SR016: system will support automated notifications
- SR017: System will support event promotion
- SR018: System will display comprehensive pet profiles
- SR019: System will support adoption application management

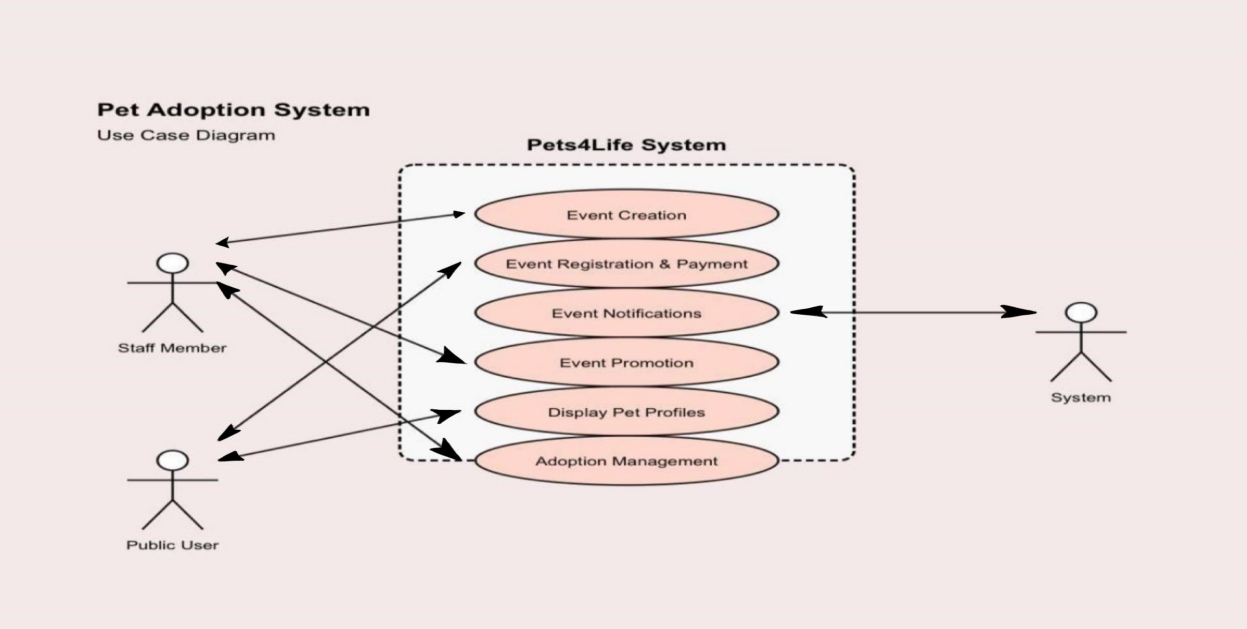
1.2.B Traceability Matrix

Pets4Life	R1	R2	R3	R4	R5	R6	R7	R8	R9	R10
UC13: Staff creates events	■									
UC14: Pet owners register for events		■								
UC15: Process event payments			■							
UC16: System sends notifications				■						
UC17: Marketing staff promotes events					■					
UC18: Public users view pet profiles						■				
UC19: Staff manages adoption applications							■			

Requirements

R1: Staff event creation functionality	R4: Automated notification system	R7: Adoption application management
R2: Public event registration system	R5: Social media integration features	R8: User account management system
R3: Payment processing capabilities	R6: Pet profile display functionality	R9: System security and performance

Use Case Diagram

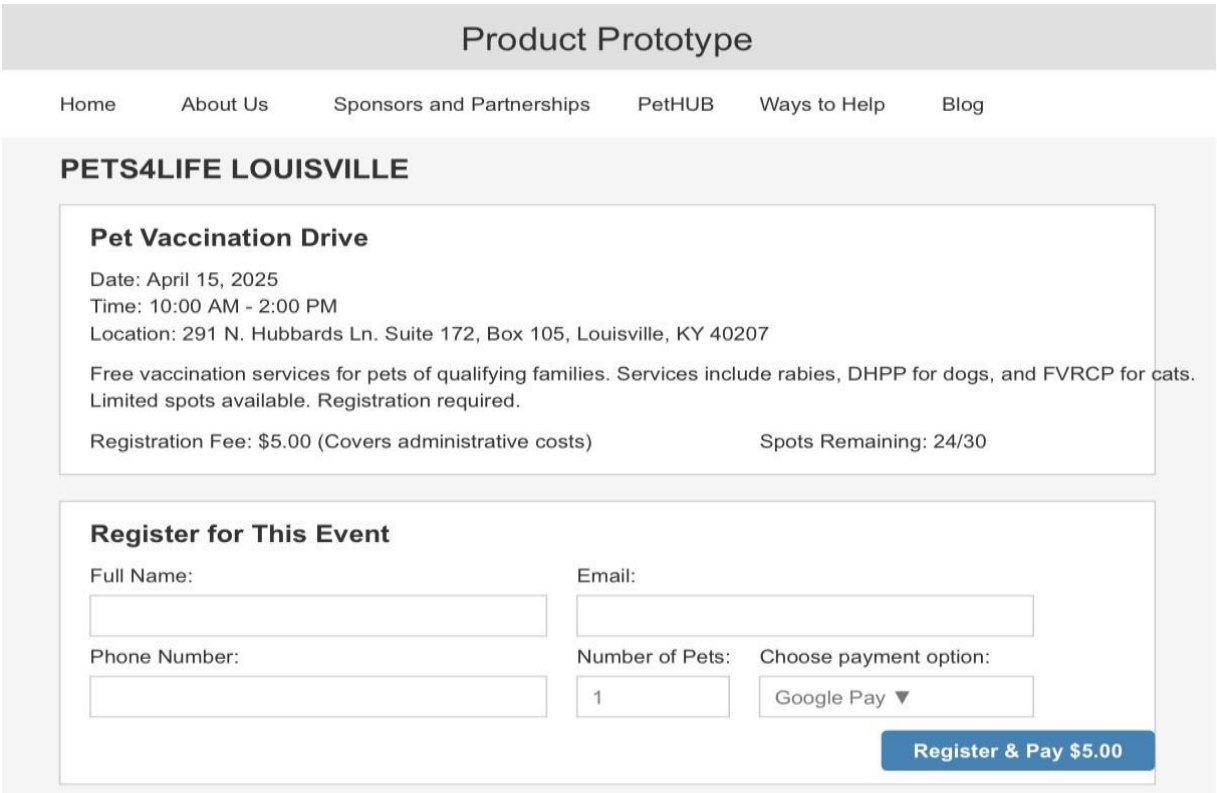


Staff members create events (Use Case 13), which can then be registered for and paid by pet owners or donors (Use Case 14). Once registered, the system automatically sends notifications (Use Case 15) to confirm and remind participants of upcoming events. Marketing staff can promote these events across social media or email campaigns (Use Case 16). Meanwhile, public users are able to view detailed pet profiles for adoption (Use Case 17). Shelter staff then manages the adoption applications for these pets (Use Case 18), approving or denying requests and updating each pet's status.

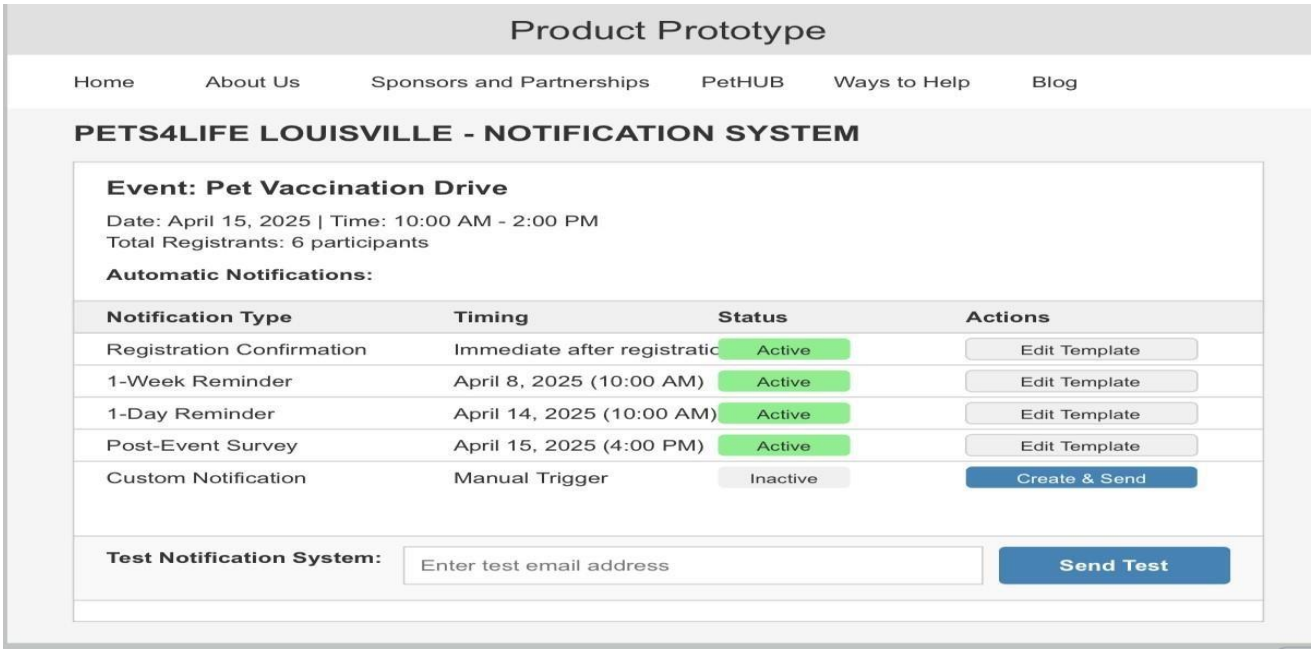
Prototypes:

Event Creation

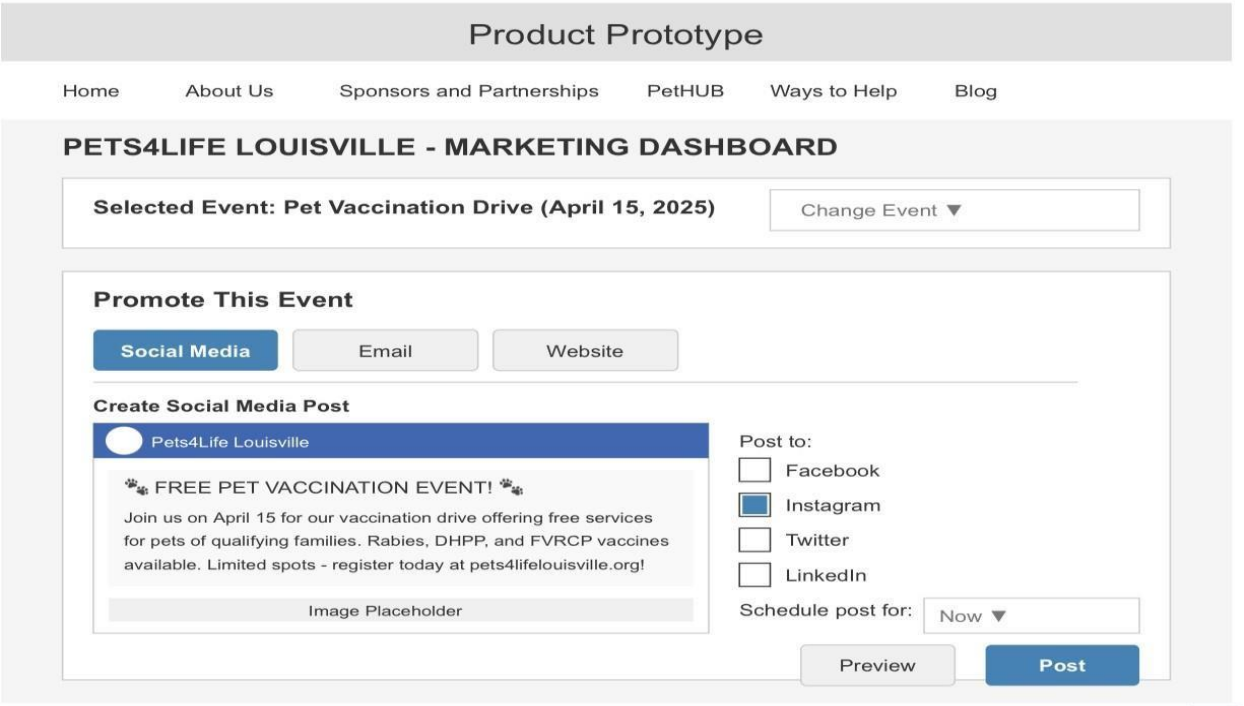
Event Registration and Payment:



Event Notification:

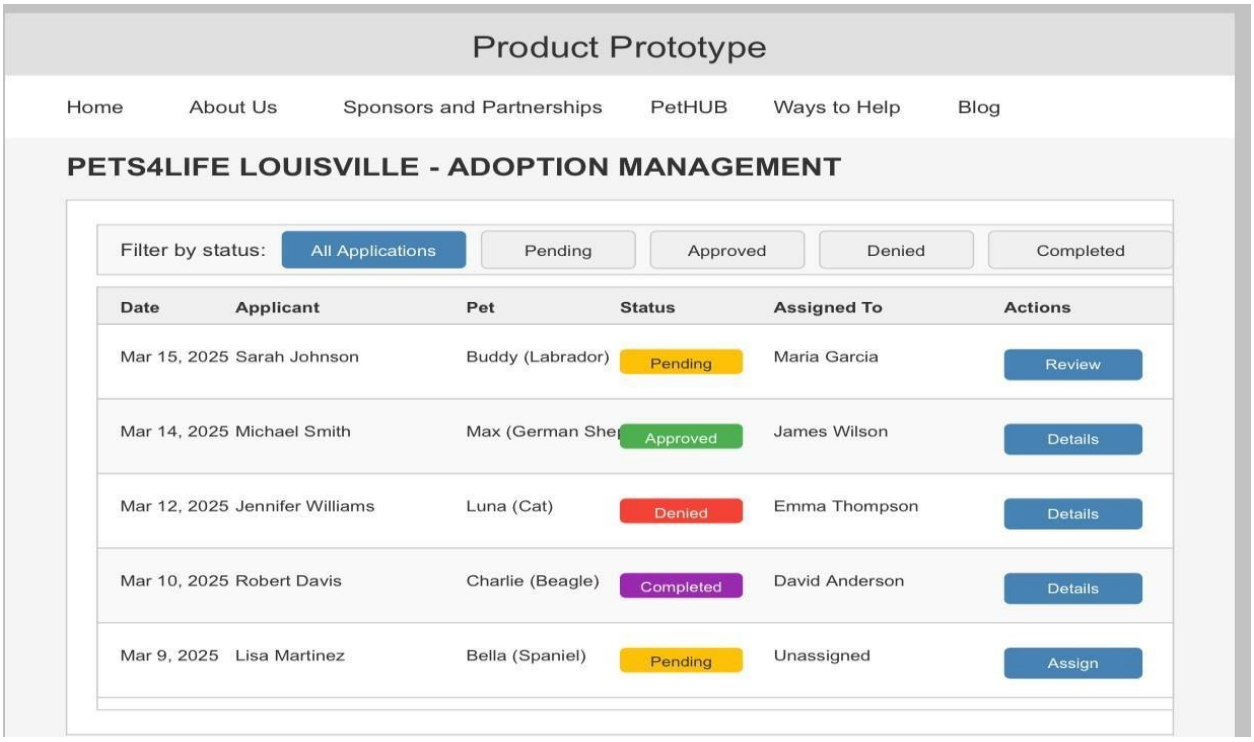


Event Promotion:



Display Pet Profiles:

Adoptiom Management :



Use Cases 20 – 23

System Requirements

ID: SR020 - The system will provide chatbot support for immediate assistance.

ID: SR021 - The system will implement a ticketing system for handling and tracking customer inquiries.

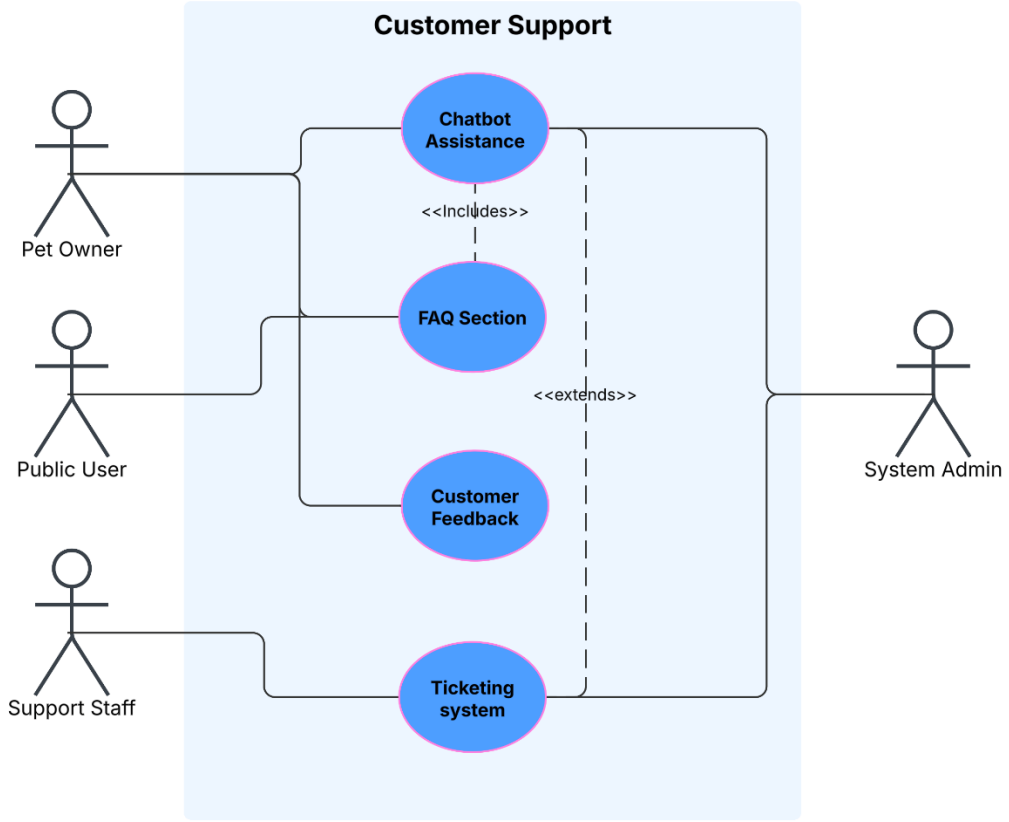
ID: SR022 - The system will include a FAQ section for common queries and concerns.

ID: SR023 - The system will offer a feedback system for customers to rate services and provide suggestions.

Traceability Matrix

	Use Case 20: Chatbot Assistance	Use Case 21: Ticketing System	Use Case 22: FAQ Section	Use Case 23: Customer Feedback	Use Case 24: Publish Content
ID: SR020 Chatbot support for immediate assistance	X				
ID: SR021 Ticketing system for handling and tracking customer inquiries		X			
ID: SR022 FAQ section for common queries and concerns			X		
ID: SR023 Feedback system for customers to rate services and provide suggestions				X	

Use Case Diagram



Use Case Narrative

There is a customer support system where users can ask questions and get self-help with their issues. The main actors that interact with the system include:

Pet Owner — Queries the chatbot for rapid responses, accesses the FAQ section, and submits feedback.

Public User – A user who mainly searches for general inquiries in the FAQ.

Support Staff – Handles escalated tickets and solves customer problems.

System Administrator — Responsible for the operation of the bot and ticketing system to ensure it is functioning.

The system consists of four use cases:

Chatbot Assistance − The automated responses chatbot works based on the queries given by the customer. It provides the appropriate information if the question aligns with the FAQ database. Otherwise, it provides an extension to the Ticketing System to be used to forward unresolved issues to Support Staff.

FAQ Section — A self-help knowledge base with frequently asked questions and answers. This infers this use case falls under Chatbot Assistance meaning the chatbot is used primarily to pull

information for the FAQ page

Ticketing System — This use case extends to create a support ticket for human intervention where the chatbot is unable to resolve any issue. This can then be assigned by Support Staff to tickets and resolved.

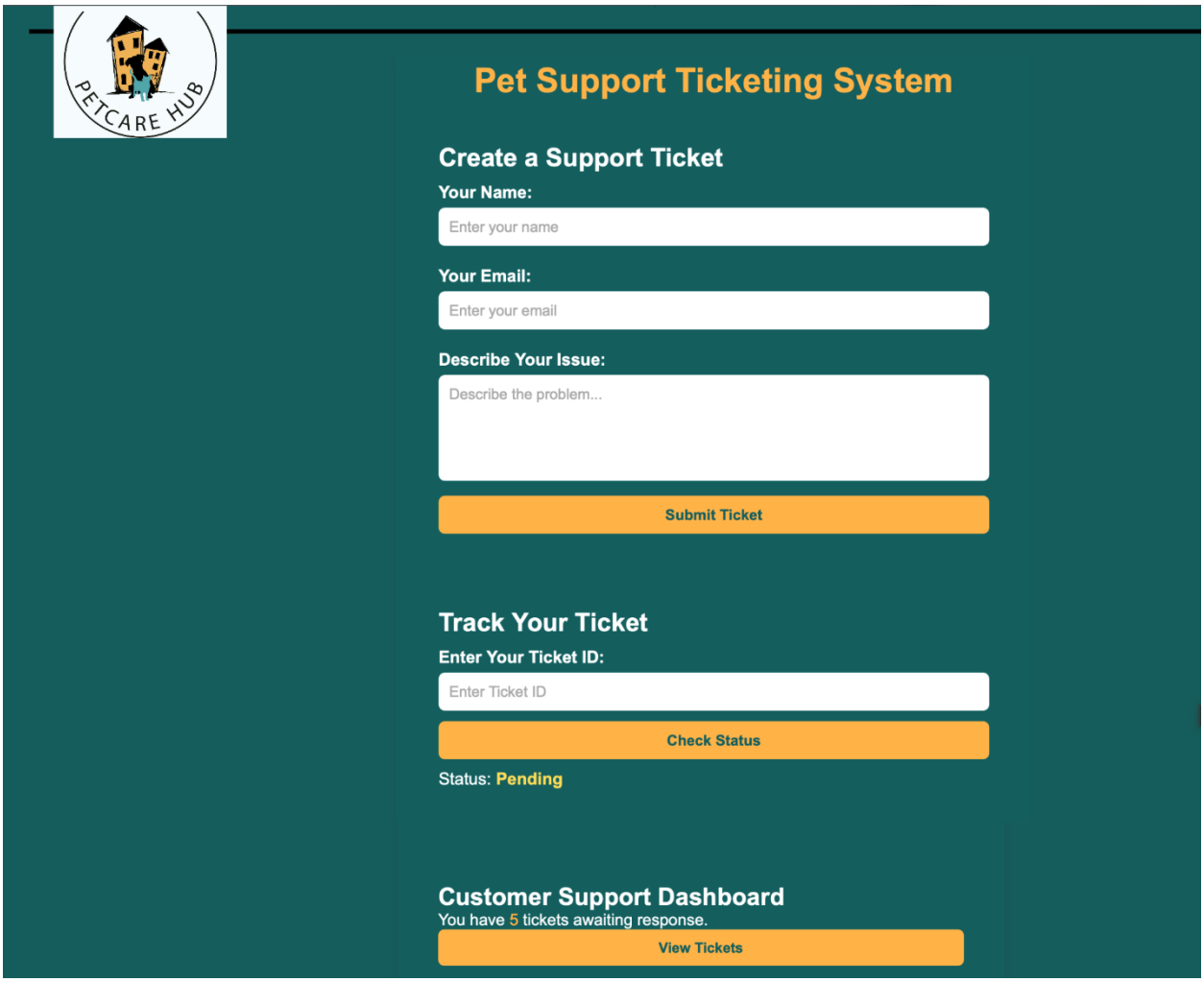
Customer Feedback – If, for some reason, there is a problem with the server, Pet Owners can raise a ticket, which will be claimed, and the issue will be resolved.

Key Relationships in the diagram:

<<includes>> relationship – The Chatbot Assistance use case includes FAQ Section as it references its knowledge base, a lot.

<<Extends>> relationship - The Chatbot Assistance use case extends the Ticketing System because an issue will only be escalated to the Ticketing System when the chatbot cannot handle the inquiry.

Prototype: Ticketing System



Use Cases 24

System Requirement

ID: SR024 – The system will provide tools for staff to post updates, blogs, and pet care information.

Traceability Matrix : Use case 24

	Use Case 24: Publish Content
ID: SR024 Tools for staff to post updates, blogs, and pet care information	X

Use Cases 26 – 30

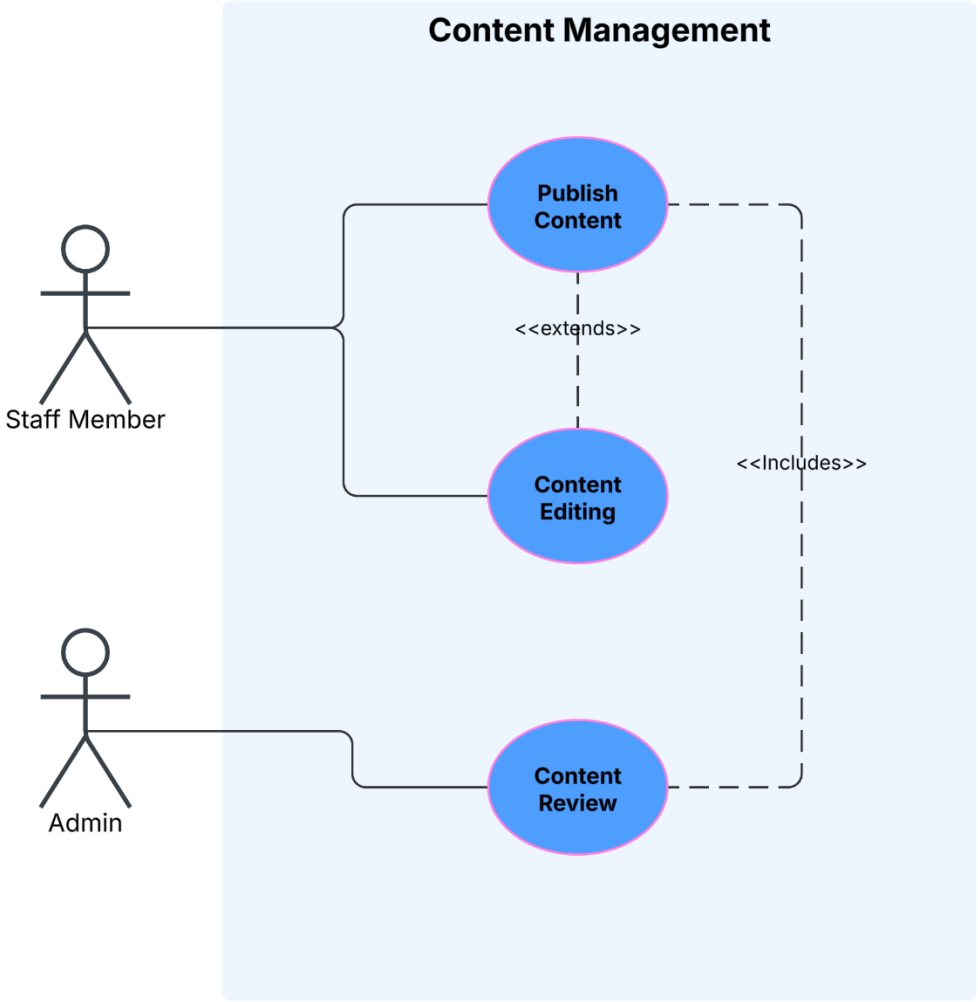
System Requirements

- **ID: SR025 – Staff member will use photo and video gallery management for event and pet highlights.**
- **ID: SR026 – System Administrator will secure management of user-generated content, including reviews and comments.**
- **ID: SR027 – Content manager will use editorial tools for content revision and approval.**
- **ID: SR028 – Administrator will access comprehensive dashboards for monitoring website traffic and user activity.**
- **ID: SR029 – Administrator will access detailed reporting tools for financial management, including donations and event revenues.**

	Use Case 26: Media Management	Use Case 27: User-Generated Content	Use Case 28: Editorial Tools	Use Case 29: Website Analytics Dashboard	Use Case 30: Financial Reports
ID: SR026 - Photo and video gallery management for event and pet highlights.	☑				
ID: SR027 - Secure management of user-generated content, including reviews and comments.		☑			
ID: SR028 - Editorial tools for content revision and approval.			☑		
ID: SR029 - Comprehensive dashboards for monitoring website traffic and user activity.				☑	
ID: SR030 - Detailed reporting tools for financial management, including donations and event revenues.					☑

Use Case Diagram

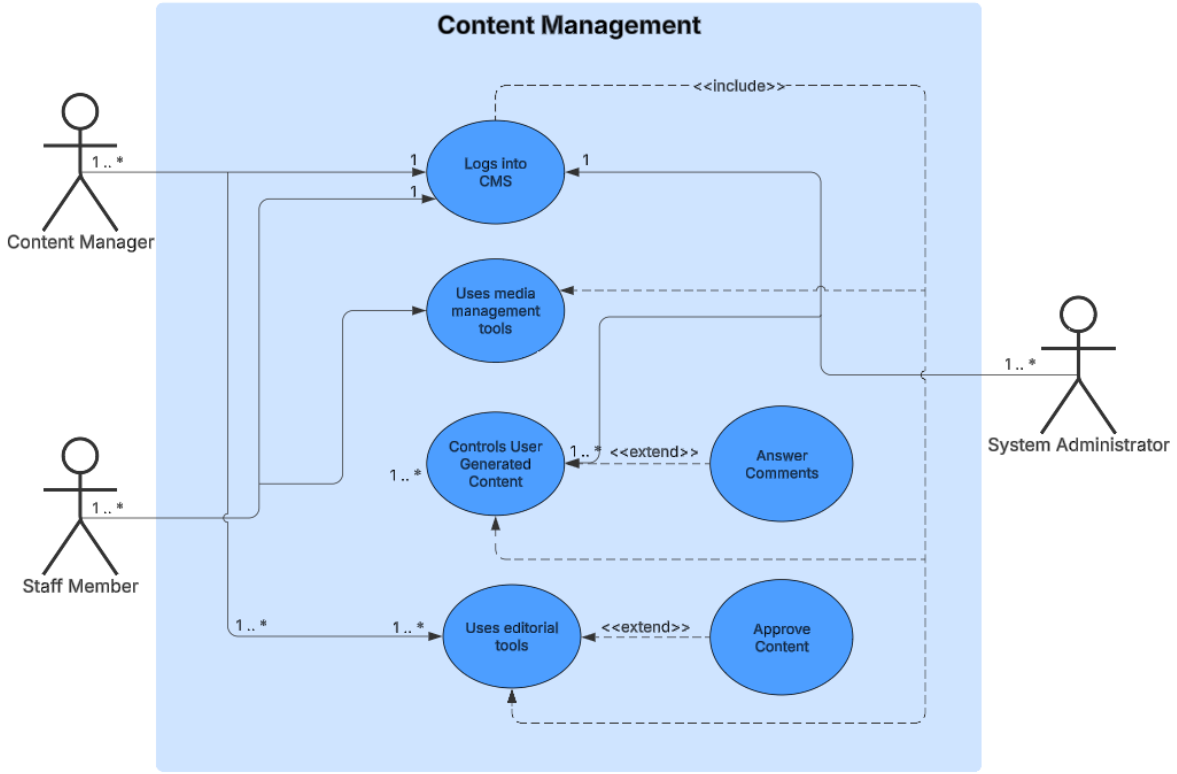
Use Case 24



Use Case Narrative

The System allows the Staff Member to publish content on the platform. The Staff Member creates and submits content, which is then published by the System. The Administrator handles any content review if approval is needed before publishing. The Publish Content use case includes the Content Editing use case, allowing the Staff Member to modify content after publication. Content Review use case extends the Publish Content use case, enabling the Administrator to approve content before it's published.

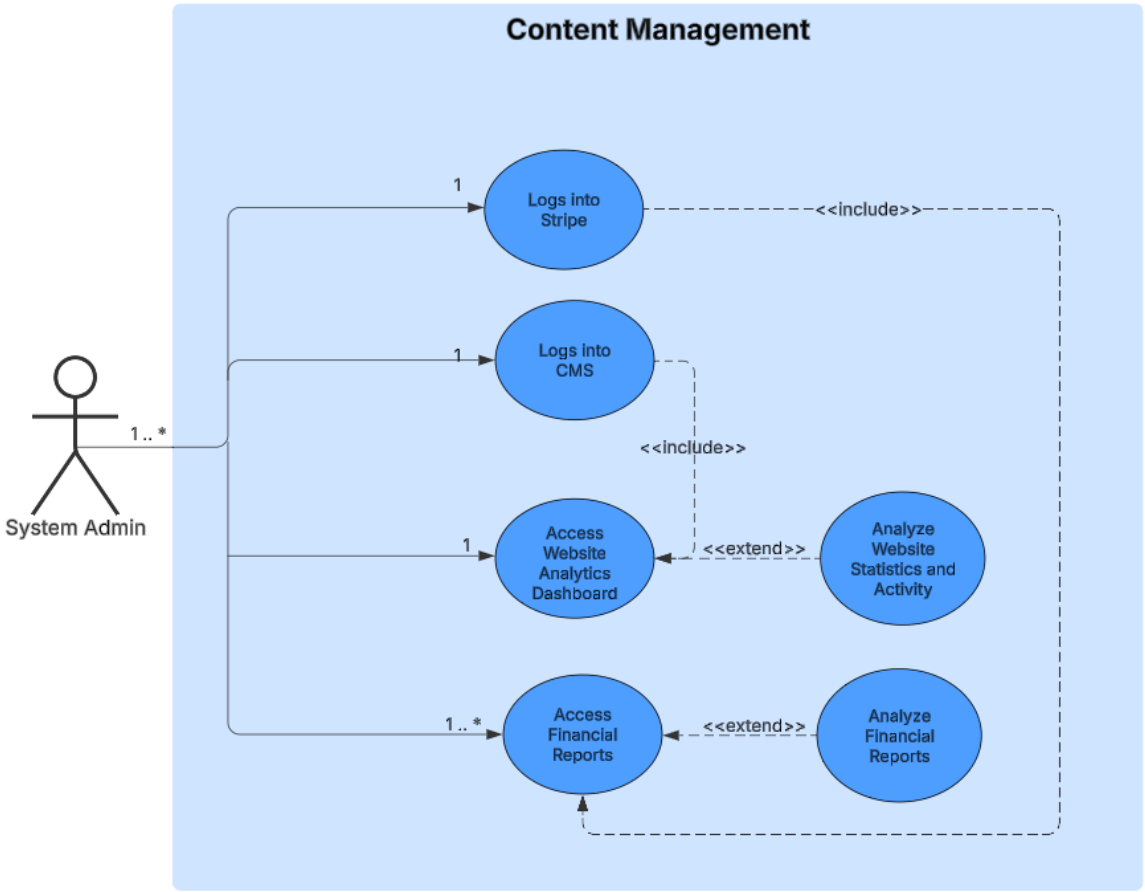
Use Cases 26-28



Use Case Narrative

The System will allow the content manager to log into the CMS and use editorial tools that extends to the use case of approving content to be published on the website. Staff members will be allowed to log into the CMS and use media management tools to control media on the website. The system will also allow system administrators to log into the CMS and control user-generated content that extends to replying to user-generated comments.

Use Cases 29-30

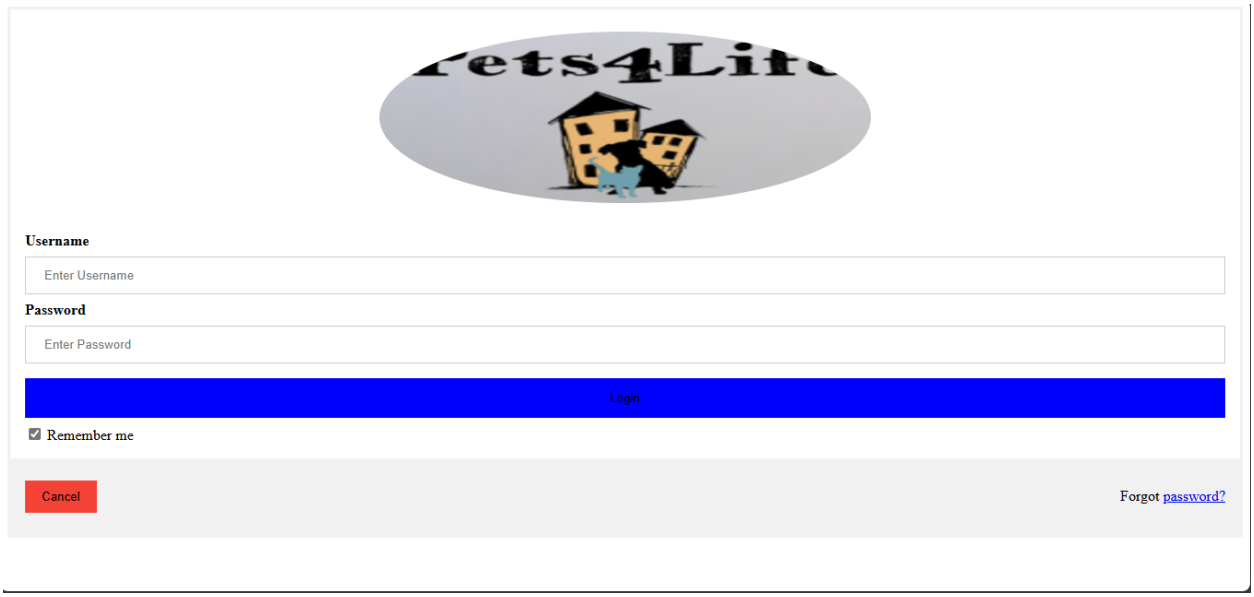


Use Case Narrative

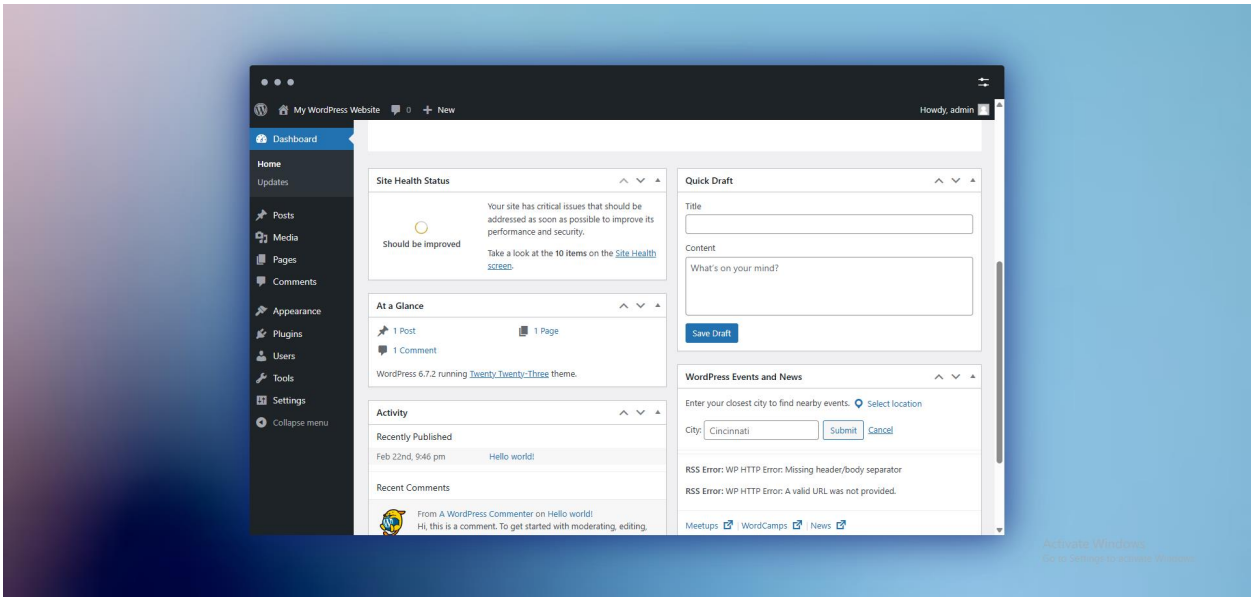
The System will allow the System administrator to log into the CMS and Stripe account. The system will also allow the system admin to access the website analytics dashboard on the CMS which extends to the use case of analyzing website statistics and activity. The system admin will also be able to access financial reports which extends to the use case of analyzing the financial information in the reports

Prototypes

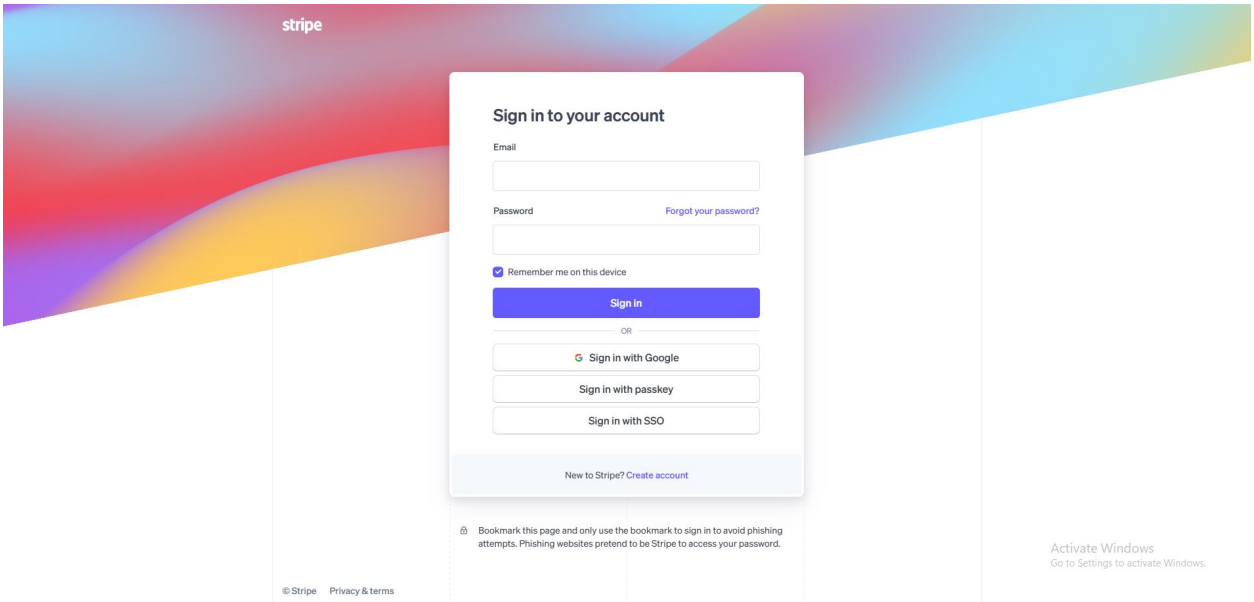
Pet4Life WordPress CMS login page



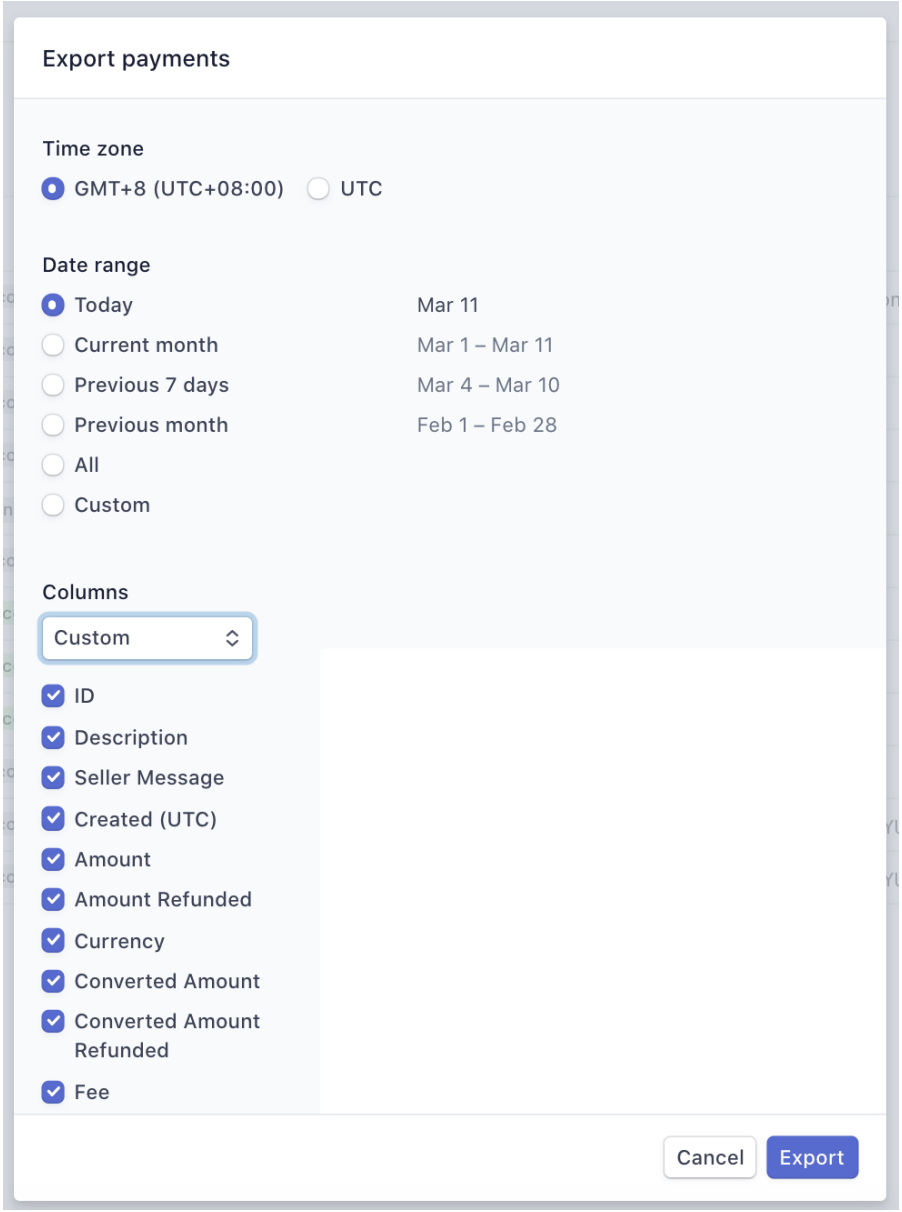
Website Analytics Dashboard



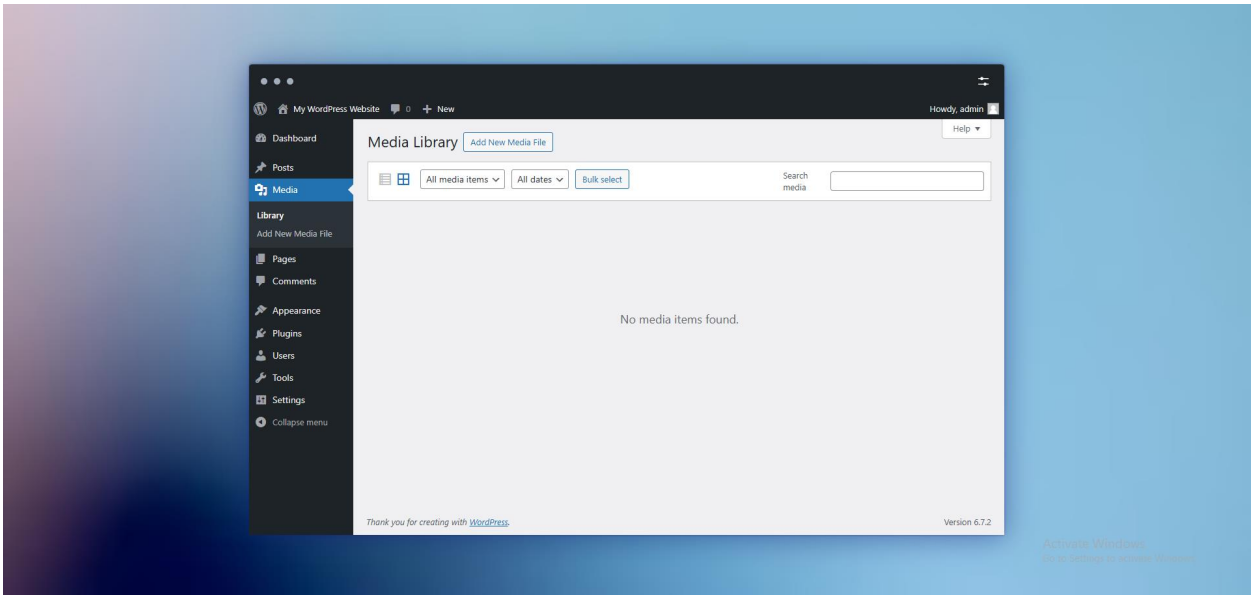
Logging into Stripe



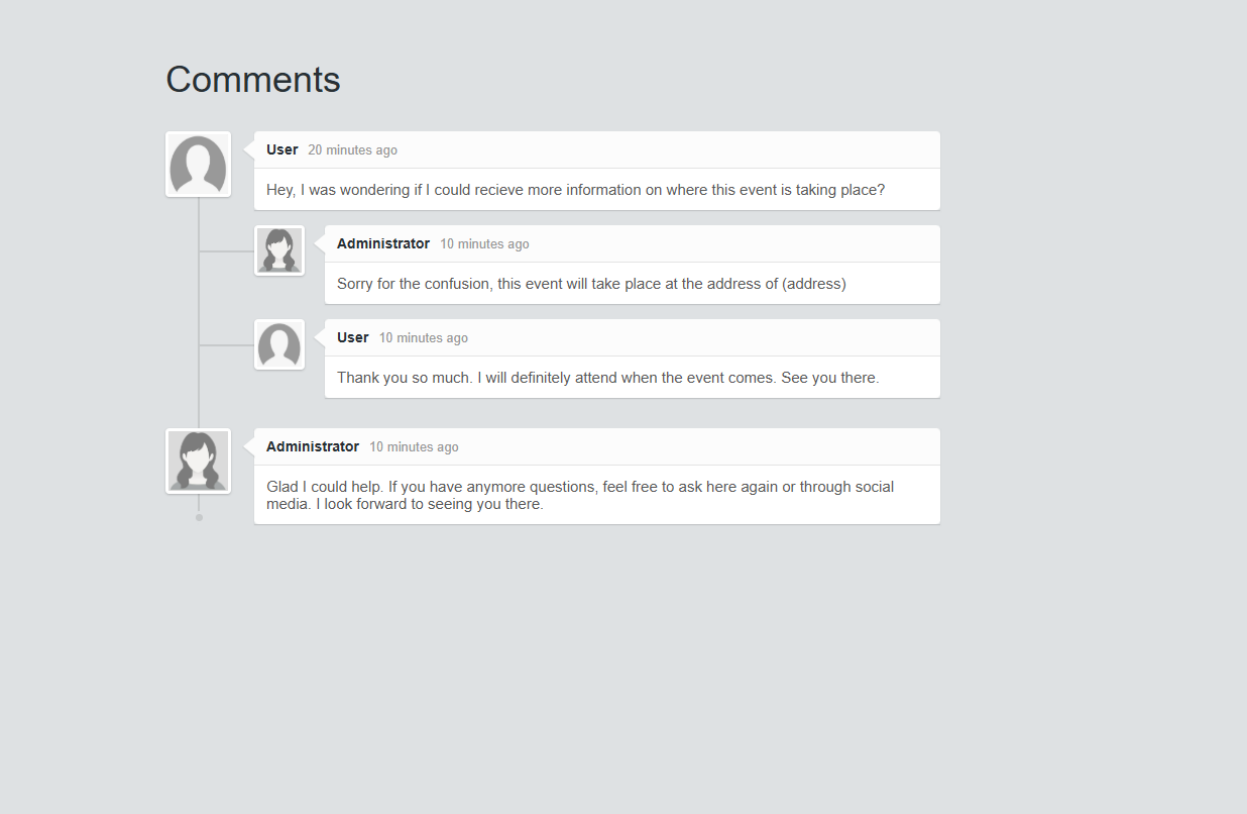
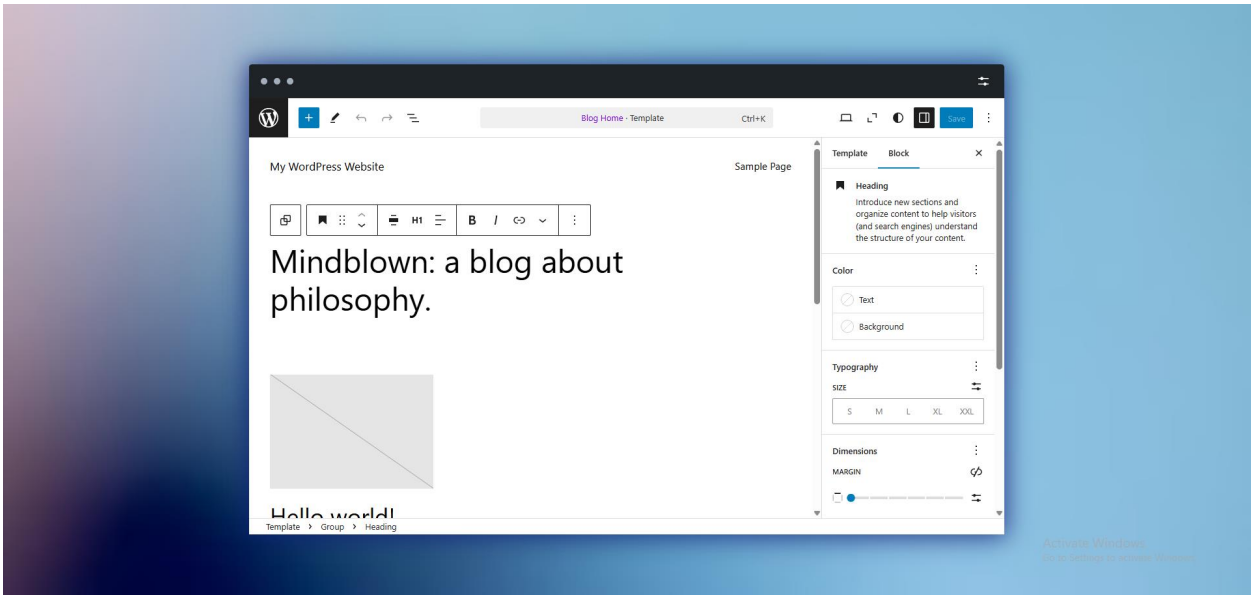
Accessing Financial reports


Export payments
Time zone
GMT+8 (UTC+08:00)
UTC
Date range
Today
Mar 11
Current month
Mar 1 – Mar 11
Previous 7 days
Mar 4 – Mar 10
Previous month
Feb 1 – Feb 28
All
Custom
Columns
Custom
ID
Description
Seller Message
Created (UTC)
Amount
Amount Refunded
Currency
Converted Amount
Converted Amount Refunded
Fee
Cancel
Export

Utilizing Media management tools



Controlling User-Generated Content



Utilizing Editorial Tools